

(Mis-)Matchmaker

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Abstract

Digital platforms act as intermediaries that control how agents are matched using detailed user data. In many matching markets, however, user and platform incentives are misaligned: users prefer fast, high-quality matches, while platforms profit from prolonged search. This paper characterizes the matching rule of a profit-maximizing platform under misaligned incentives. I show that two commonly studied matching rules, random matching and positive assortative matching (PAM), can be suboptimal. When surplus extraction through prices is constrained, due to advertising-based business models or pricing frictions, the platform strategically delays matches, generating excessive search and inefficient mismatch relative to the social optimum.

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1. INTRODUCTION

Many digital platforms act as intermediaries that control how agents are matched. By conditioning matching on detailed user data, these platforms can shape both the quality and timing of matches. While improved information has the potential to increase matching efficiency, the incentives of users and platforms are often misaligned: users prefer fast, high-quality matches, whereas platforms profit from prolonged search. This paper investigates the impact of a platform with detailed user data on the resulting speed and assortativity of matching in the society. It highlights a novel source of mismatching and delay: profit-driven, purposeful distortion of the matching rule by the platform.

Online dating and job platforms are prominent examples of such intermediaries, where matching is mediated by algorithms and users incur costs while searching. In fact, online dating has become the most common way to meet potential partners in recent years, and for more than a decade, job searches have been conducted predominantly through such online platforms (Rosenfeld et al., 2019; Kircher, 2022).¹ In both dating and labor markets, platforms predominantly rely on advertising or per-period subscription fees, i.e., business models whose revenues increase with user engagement rather than match success. Moreover, algorithmic matching is rapidly extending into new domains with the help of AI: Microsoft has launched AI agents for patient-to-clinical-trial matching in healthcare,² while LinkedIn’s AI hiring assistant now autonomously sources and screens job candidates.³ The incentives to distort matching resulting from a misalignment in incentives are thus likely to become more prevalent in the future.

This paper shows that the two predominant search protocols used to study centralized and decentralized matching markets — the positive assortative matching (PAM) rule and random matching — can be suboptimal for the platform. Random matching is generically suboptimal whenever the platform has access to user data. PAM is suboptimal whenever surplus extraction through prices is constrained, as under advertising-based business models or price ceilings. Instead, the platform strategically distorts matching to induce longer search. This creates two distinct inefficiencies: excessive search and inefficient mismatches, both of which generate welfare losses relative to the social optimum.

¹Similar intermediation structures arise in ride-sharing markets, where platforms control driver-rider assignment and have incentives to manage driver positioning to increase labor supply rather than minimize wait times (Chen et al., 2024). Social media platforms match users to content and are documented to prioritize engagement over user satisfaction (Milli et al., 2025). In financial markets, retail brokers route investor orders to marketmakers under payment-for-order-flow arrangements, creating incentives to favor wholesalers offering the highest broker commissions over those offering the best execution for clients (Ernst and Spatt, 2023).

²See <https://news.microsoft.com/source/2024/10/10/microsoft-expands-ai-capabilities-to-shape-a-healthier-future>, last accessed 19.05.2026.

³See <https://news.linkedin.com/2025/hiring-assistant-globally-available>, last accessed 19.05.2026.

I study these distortions in a model of a monopoly platform organizing a two-sided matching market in which users search for a partner on the opposite side. The platform commits to a matching rule that determines the probability that two users, each characterized by a vertical type, will meet. Focusing on the most prominent business models, the platform commits to either an amount of advertising or a payment per period in which the user is active. In either case, spending their time searching is costly for users. To attract and keep users' attention, the platform recommends users a match in each period. Upon receiving a recommendation, users simultaneously decide whether to accept or reject the proposed match; after rejecting, they continue to search. The analysis focuses on steady states, in which the inflow of new agents must equal the outflow under the platform's matching rule.

Characterizing the platform's optimal matching rule requires solving a problem that is highly non-linear in the original formulation. Fixing search costs, the platform's problem is to choose matching probabilities conditional on each users' type subject to participation constraints regarding the users' decision to join the platform, incentive constraints on the users' acceptance decisions, feasibility constraints on the matching mechanism as well as steady-state constraints. The paper's main methodological contribution is a linear programming reformulation of this dynamic matching problem with an endogenous meeting rule. The approach applies to environments in which payoffs depend linearly on the mass of matches, and provides a tool for a broad class of platform design, search, and assignment problems.

In the linear programming problem, the platform chooses masses of recommended and matched pairs using the facts that: (i) the objective function is linear in steady-state masses, and (ii) the constraints are linear in the mass of recommended and matched pairs by using appropriate transformations. In the general setting for any given finite set of users' types, I establish existence of an optimal solution to the platform's problem using the auxiliary problem. Based on the reformulation, I prove two main results about the platform's matching rule. First, random matching is generically suboptimal whenever the platform has access to user data: by conditioning meeting probabilities on types, the platform can always improve upon type-independent matching, implying that data is strictly valuable. Second, PAM is suboptimal whenever surplus extraction through prices is constrained.

The result offers an explanation for why platforms rely on business models with misaligned incentives. With full pricing power, the platform would charge type-dependent fees and implement PAM, extracting the entire match surplus in a single period. When surplus extraction is constrained, e.g., because users are reluctant to pay upfront but are willing to consume advertising, an advertising-based model can be more profitable if

targeted advertising is sufficiently efficient.⁴ More generally, when pricing is constrained, the platform substitutes rent extraction via prices for rent extraction via search. Under price ceilings, it strategically lowers match quality to induce longer search, and thereby increases the payments it collects. Under uniform pricing, the platform frequently recommends and fosters mismatches to prolong search and thus, screens users through search duration rather than prices.

The special case with two types on each side and symmetric inflows delivers a full characterization of matching outcomes and illustrates the main implications for the speed and quality of matching. The profit-maximizing platform may not implement the socially optimal positive assortative outcome. Instead, it creates two economic inefficiencies. The first is delay: the platform recommends low types that are rejected to high types (“*mis-meetings*”), inducing additional periods of search and thereby increasing profit. The second is deliberate mismatch: the platform induces agents to permanently leave the market in inefficient pairs rather than continue searching for their assortative match. By inducing high types to accept low types (“*mismatches*”), the platform relaxes feasibility constraints for high types and extracts additional rent. Both distortions have implications for real-world markets such as dating and labor markets: in labor markets, mismatch reduces productivity and contributes to long-term unemployment (Şahin et al., 2014; McGowan and Andrews, 2015), while in marriage markets, sorting shapes income inequality and match quality affects well-being (Lee, 2016; Robles et al., 2014; Sharabi and Dorrance-Hall, 2024).

The paper is organized as follows. Section 2 discusses related literature. The model is presented in Section 3. Section 4 characterizes the platform’s problem, in which Section 4.1 analyzes agents’ strategies and Section 4.2 derives the linear-programming reformulation. The main results are in Section 4.3, which establishes the suboptimality of PAM and random matching. The analysis is applied to binary types in Section 4.4. Lastly, Section 5 discusses competition and network effects, and Section 6 concludes.

2. RELATED LITERATURE

This paper connects the literature on matching-and-search theory and platform markets. In contrast to the existing literature, I consider the profit-maximizing incentives of a matchmaker when agents are vertically differentiated and characterize the matching rule and resulting matching outcome.

⁴Advertising-based models play a key role in online markets, including both fully ad-supported and “freemium” business models. Freemium refers to business models, where users can use a basic service for free in exchange for consuming ads, but need to pay a fee to use the premium service (without ads).

Matching-and-Search Theory The vast literature on search-and-matching models, see for instance Burdett and Coles (1999), Eeckhout (1999), and Smith (2006), provides insights into the functioning of decentralized markets in which agents meet at “random”.⁵ These matching models with heterogeneous agents build the foundation to investigate sorting and mismatch in markets such as labor and marriage markets when search frictions are present; for a recent overview see Lauermann and Nöldeke (2025). In line with these models, agents in my model have vertical preferences that result in a unique stable matching. I follow Lauermann and Nöldeke (2014) and suppose that types are finite. The model at hand crucially departs from the literature on decentralized matching, which assumes that agents meet according to a random matching technology, by explicitly accounting for the design of the matching rule. With increasing access to user data about preferences and machine-learning tools, matching platforms can design their own recommendation and matching algorithms to maximize profits. While many platforms do not disclose the specifics of their matching algorithms, it is evident that their algorithms are far more sophisticated than random matching.⁶

In the search-and-matching literature, search intensity is often treated as exogenous. A notable exception is Bloch and Ryder (2000), who study a matchmaker facilitating search between agents with vertical types. Bloch and Ryder (2000) analyze two regimes: random matching and a matchmaker implementing identity matching (the analogue of PAM in my model) under a commission fee or a uniform participation fee. In their setting, the matchmaker improves efficiency relative to decentralized search by implementing PAM. In contrast, I show that PAM is suboptimal when pricing is constrained.

The question of how to design the matching rule is related to the literature on centralized matching as pioneered by Gale and Shapley (1962), Roth (1982) and Roth and Sotomayor (1992), which studies match quality and implementation of efficient matching rules in two-sided markets.⁷ The literature considers properties such as stability, strategy-proofness and Pareto efficiency of the matching rule. In contrast, I do not consider a social planner but a profit-maximizing intermediary given different business models.

Search problems are widely studied not only at the individual level but also as tools for understanding aggregate labor market dynamics. Early articles include Pissarides (1985), Mortensen and Pissarides (1994), and Mortensen and Pissarides (1999), which

⁵The aforementioned literature assumes that agents have non-transferable utility. Search-and-matching models with transferable utility have been analyzed, for example, by Becker (1973, 1974) and Shimer and Smith (2000).

⁶Dating platforms such as Tinder or bumble provide a general description of their algorithm, see for example <https://www.help.tinder.com/hc/en-us/articles/7606685697037-Powering-Tinder-The-Method-Behind-Our-Matching>, whereas the dating platform “Hinge” claims to use the Gale-Shapley algorithm designed to find stable matchings.

⁷The literature on matching in two-sided markets can be divided into centralized and decentralized matching (see Echenique et al. (2023) for a recent overview).

focus on wage bargaining, unemployment dynamics and on-the-job search when agents are ex-ante homogeneous. Dolado et al. (2009) introduces heterogeneous types of workers and firms into job search models, which are also crucial in my model. A recent treatment on how job search has changed in the digital era is provided by Kircher (2022).

Platform Markets The second strand is the literature that studies platforms and two-sided markets. Central to that literature is the presence of network effects and how these shape the incentives and price setting of a platform that enables the interaction between two groups (Caillaud and Jullien, 2003; Rochet and Tirole, 2003, 2006). As a result, in most models agents are assumed to care only about the number of matches instead of match quality. With the emergence of digital matchmakers, the literature extended to analyzing (customized) matching on platforms with a focus on the interaction between pricing and matching efficiency (Damiano and Li, 2007; Damiano and Hao, 2008), price discrimination (Gomes and Pavan, 2016, 2024), and auctions (Johnson, 2013; Fershtman and Pavan, 2022), all abstracting from search frictions and dynamics. In my model, the platform designs the matching rule in its matching market, but in contrast to the aforementioned articles, the platform has an incentive to not implement the socially optimal matching rule. Within the analysis of digital matchmakers, Halaburda et al. (2018) and Antler et al. (2023, 2026) focus on applications to dating platforms.

Most closely related is Antler et al. (2026), who study a matchmaker’s incentives in a model with horizontally differentiated types, where match quality is revealed gradually as agents spend time together off the platform. The platform receives a signal about prospective fit and uses it to screen matches above a threshold, while agents within the screened pool meet at random. The authors investigate whether a platform adopts new technologies that can improve the speed of matching and screening. They find that the platform always benefits from faster matching, whereas investing in match quality through better screening is profitable when the platform can optimally readjust its fee, but not under an advertising-based business model. The main difference lies in the modeling of the matching technology. Antler et al. (2026) restrict attention to a truncated random matching rule and ask whether the platform invests in improving it; in contrast, I take the platform’s information as given and solve for the profit-maximizing matching rule. This reveals a distinct source of inefficiency: whereas faster matching is always beneficial in their framework, I show that a platform with detailed user data may deliberately delay matching and induce mismatches to extract rents.

Within the platform literature, models on a (monopoly) platform intermediating consumer search are closely related (Hagiu and Jullien, 2011; Eliaz and Spiegler, 2016; Nocke and Rey, 2024). Hagiu and Jullien (2011) provide a rationale for intermediaries to divert

search of their consumers away from preferred stores. Although the insight is closely related to the mismatching incentive in my model, the (one-sided) market in Hagiu and Jullien (2011) does not include the strategic component on the other side as stores would never reject a consumer willing to buy. Hence, there is no analogue to my finding that the platform prolongs search of lower types by recommending them to higher types knowing that they will reject those lower types. Finally, my model of a two-sided matching market offers insights into the allocative inefficiency and the length of search for labor and dating markets intermediated by matching platforms.

3. MODEL

A monopolist platform organizes a matching market in which a continuum of agents from two sides, $k = A, B$, search for a partner from the opposite side. The market operates in discrete time with an infinite horizon. I focus on steady state analysis. In slight abuse of notation, I therefore suppress time indices whenever it does not lead to confusion.

Agents An agent of each side is characterized by a type $\theta_i^k \in \Theta^k$, with $\Theta^k = \{\theta_1^k, \theta_2^k, \dots, \theta_{N^k}^k\}$ finite. At the beginning of each period, an agent θ_i^k decides whether to enter the market or to exit and take outside option ω_i^k . An agent that participates in the market becomes inactive with an exogenous probability $\delta > 0$ and also leaves the search process. The platform charges an active agent of type θ_i^k a search cost s_i^k . Then, each active agent receives a single recommendation from the platform. After receiving a recommendation, two agents who meet observe each other's type and simultaneously decide whether to accept or reject the other agent. The following payoffs are realized based on their actions in the current period: (i) mutual acceptance yields a match utility of $u(\theta_i^k, \theta_j^{-k}) = \theta_i^k \theta_j^{-k}$,⁸ and (ii) (one-sided) rejection yields a utility of zero in the current period. After a rejection, an agent can continue searching in the next period.

Agents are assumed to use time- and history-independent strategies. A pair of functions $\sigma_k : \Theta^k \times \Theta^{-k} \rightarrow [0, 1]$ and $\sigma_{-k} : \Theta^k \times \Theta^{-k} \rightarrow [0, 1]$ describe the acceptance strategies, where $0 \leq \sigma_k(\theta_i^k, \theta_j^{-k}) \leq 1$ is the probability that an agent of type θ_i^k on side k accepts a match with type θ_j^{-k} on the other side. The function $\eta_i^k : (\theta_i^k, \omega_i^k) \rightarrow [0, 1]$ describes the participation strategy of an agent of type θ_i^k with outside option ω_i^k . In other words, without loss of generality, I focus on strategies where identical agents, active on the same side of the market and of the same type, use the same acceptance and

⁸Supermodularity of the match utility ensures that PAM is the unique social optimum and here imposes symmetric payoffs across sides. The weaker condition that match utility is monotone in the partner's type, however, suffices for PAM to be socially optimal. Under submodularity, the social optimum is negative assortative matching (NAM), and the platform's incentives to distort away from NAM can be characterized analogously.

participation strategy. Then,

$$\alpha(\theta_i^k, \theta_j^{-k}) = \sigma_k(\theta_i^k, \theta_j^{-k}) \cdot \sigma_{-k}(\theta_i^k, \theta_j^{-k})$$

denotes the probability of a mutual acceptance by type θ_i^k and θ_j^{-k} .

Matching A matching mechanism $\mathcal{M} := \{\phi^k(\cdot)\}_{k=A,B}$ consists of (potentially stochastic) matching rules $\phi^k(\cdot)$. Let $\hat{\Theta}^k$ be the set of participating types from side $k = A, B$. For $\theta_i^k \in \hat{\Theta}^k$, $\phi^k(\cdot|\theta_i^k) \in \Delta(\hat{\Theta}^{-k} \cup \omega_i^k)$, which is a probability measure over $\hat{\Theta}^{-k} \cup \omega_i^k$. Intuitively, this describes the probability of meeting the various types of the opposing side as well as the outside option. Any $\theta_i^k \in \Theta^k \setminus \hat{\Theta}^k$ who does not participate is assumed to be meet their outside option with probability one, $\phi(\omega_i^k|\theta_i^k) = 1$. Denote the mass of agents of type θ_i^k on side k by $f(\theta_i^k)$. Matching mechanism $\mathcal{M}$ induces a distribution of matched pairs M

$$\left(\begin{pmatrix} f(\theta_1^k) \\ \vdots \\ f(\theta_{N^k}^k) \end{pmatrix}, \begin{pmatrix} f(\theta_1^{-k}) \\ \vdots \\ f(\theta_{N^{-k}}^{-k}) \end{pmatrix} \right) \mapsto \begin{pmatrix} \Phi(\theta_1^k, \theta_1^{-k}) & \cdots & \Phi(\theta_1^k, \theta_{N^{-k}}^{-k}) \\ \vdots & & \vdots \\ \Phi(\theta_{N^k}^k, \theta_1^{-k}) & \cdots & \Phi(\theta_{N^k}^k, \theta_{N^{-k}}^{-k}) \end{pmatrix} \equiv M.$$

An entry of matrix M is the mass of agents that are recommended to each other under matching mechanism $\mathcal{M}$ and is given by

$$\Phi(\theta_i^k, \theta_j^{-k}) = f(\theta_i^k)\phi(\theta_j^{-k}|\theta_i^k) = f(\theta_j^{-k})\phi(\theta_i^k|\theta_j^{-k}),$$

where the masses are symmetric, i.e. the mass of agents of type θ_i^k on side k being matched to agents of type θ_j^{-k} on side $-k$ is equal to the mass of agents of type θ_j^{-k} on side $-k$ being matched to type θ_i^k on side k : $\Phi(\theta_i^k, \theta_j^{-k}) = \Phi(\theta_j^{-k}, \theta_i^k)$. Under matching mechanism $\mathcal{M}$, the mass of agents of type θ_i^k that are unmatched, i.e. do not receive a recommendation in a given period, is

$$\Phi(\theta_i^k, \omega_i^k) = f(\theta_i^k) - \sum_{\theta_j^{-k} \in \Theta^{-k}} \Phi(\theta_i^k, \theta_j^{-k}).$$

To capture the idea that the platform can only generate revenue by keeping users' attention, I impose the following assumption.

Assumption 1. *Let $\hat{k}$ be the short side of the market. For each agent on side $\hat{k}$, $\phi(\omega_i^{\hat{k}}|\theta_i^{\hat{k}}) = 0$.*

Feasibility of the matching rule can be expressed in terms of matched-pair masses.

Definition 1. A matching mechanism $\mathcal{M}$ is feasible if

$$\sum_{\theta_j^{-k} \in \Theta^{-k}} \Phi(\theta_i^k, \theta_j^{-k}) + \mathbf{1}_{k=\hat{k}} \Phi(\theta_i^k, \omega_i^k) = \eta_i^k f(\theta_i^k), \forall \theta_i^k \in \Theta^k, k = A, B. \quad (1)$$

Timing and Population Dynamics At the beginning of a period t , agents who did not find a match in the last period arrive and a (time-invariant) inflow of new agents of type θ_i^k given by the mass $\{\beta_i^k\}_i^{k=A,B}$ enters the platform. Agents decide whether to participate on the platforms. Those who decide to participate become inactive with probability δ , while active agents are matched according to matching mechanism $\mathcal{M}$ resulting in matrix M_t . Based on their recommended match, agents make their acceptance decision resulting in mutual acceptance probabilities $\{\alpha_t(\theta_i^k, \theta_j^{-k})\}_{ij}$. At the end of the period, agents that mutually accepted each other exit in pairs. The total outflow of agents is then given by pairs that exit together in a match, agents that become inactive with probability δ and agents that decided not to participate.

Platform The platform commits to a matching mechanism $\mathcal{M} := \{\phi^k(\cdot)\}_k$. To capture the two most prominent business models, I suppose that the platform either commits to an extent of advertising or a given payment per period. Formally, this choice induces the type-dependent search cost s_i^k while generating revenue per search of type θ_i^k of $\nu(s_i^k)$. In case of payments, $\nu(s_i^k)$ is the identity function. In case of advertisements, $\nu(s_i^k)$ is an increasing and strictly concave function of the search costs, which for example captures the intuition that the agents' disutility of advertising is convex in the number of ads shown while the platform's profit is constant per ad. Let $s_i^k \in [0, \bar{u}]$, where $\bar{u}$ is the maximum match utility that the highest type can achieve on the platform. The platform discounts future profits according to ρ and thus maximizes

$$\Pi = \sum_{k=A,B} \sum_{\theta_i^k \in \Theta^k} \frac{(1-\delta)\eta_i^k}{1-\rho} \nu(s_i^k) f(\theta_i^k).$$

Equilibrium Concept The model focuses on a steady state analysis in which the market is balanced: that is, the inflow of agents is equal to the outflow of agents under matching mechanism $\mathcal{M}$. Formally:

Definition 2. (Steady State) For given matching mechanism $\mathcal{M}$, a *steady state* is a tuple $(f(\theta_i^k), \alpha(\theta_i^k, \theta_j^{-k}), \eta_i^k)_{ij}^k$ that satisfies

$$\beta_i^k = f(\theta_i^k) \left[(1 - \eta_i^k) + \eta_i^k \left(\delta + (1 - \delta) \sum_{\theta_j^{-k} \in \Theta^{-k}} \alpha(\theta_i^k, \theta_j^{-k}) \phi(\theta_j^{-k} | \theta_i^k) \right) \right], \quad (2)$$

for all $\theta_i^k \in \Theta^k, k = A, B$. The left-hand side describes the inflow of agents of type θ_i^k , where the right-hand side is the outflow. The outflow is the mass of type θ_i^k agents times the probability that agents do not participate plus the probability of becoming inactive or exiting in a match.

A steady state is an equilibrium if the following is satisfied.

Definition 3. (Equilibrium) A steady state is an equilibrium if — given that agents anticipate other agents’ strategies correctly — the profile of stationary strategies (σ, η) satisfies:

1. Agents accept a match if and only if the match yields a higher payoff than the expected utility from continuing to search.
2. Agents participate if and only if the expected utility from participating yields a higher payoff than their outside option.

Under the usual Nash assumption of correctly anticipating other players’ strategies, the definition captures that agents maximize expected utility with respect to their acceptance strategy implicitly ruling out the case that a valuable pair is rejected because everyone is certain that their partner rejects.⁹ The third part captures that agents maximize expected utility when deciding to participate on the platform.

3.1 DISCUSSION OF ASSUMPTIONS

I now turn to the key assumptions of the model. The model assumes that the platform is a monopolist in the matching market. In two-sided markets, platforms often have large market power, because joining a new platform is valuable only if others join (Rochet and Tirole, 2003, 2006). My monopoly setup is a simple setting capturing such market power. I return to this assumption in Section 5, where I consider a monopolist facing a competitive fringe.¹⁰

Search frictions are modeled by introducing the exogenous exit probability δ . Following a literal interpretation, δ is the probability with which agents become inactive, i.e. the probability that an agent finds a job or a partner offline through other means.

⁹This allows the current match partner to tremble with small probability. Alternatively, acceptance decisions could be made sequential in which case agents would have to accept a valuable match.

¹⁰For example, the dating market is highly concentrated, with Match Group Inc. owning many of the most popular platforms: Tinder, Hinge, PlentyofFish, Match, OkCupid, etc. (see <https://www.bamsec.com/filing/89110323000114?cik=891103>). For simplicity, I assume the dominant owner operates only one platform. Competing platforms are often highly differentiated, catering to niche groups (e.g., religious users), and recent evidence from Dertwinkel-Kalt et al. (2024) shows that even close competitors like Tinder and Bumble are perceived by users as nearly independent rather than substitutes.

More generally, δ can be thought of as modeling the force that leads agents to discount the future, which makes delayed matching more costly. Additionally, agents incur additive search costs s_i^k in each period, which are designed by the platform. They either represent the nuisance costs from advertising as, for example, in Anderson and Coate (2005), which are positively related to the advertising intensity, or the search fee that the platform charges periodically.

The model examines two prevalent business models: (targeted) advertising and search fees. Many platforms adopt the former by monetizing user attention through selling advertising slots to firms. In return for users’ attention, the platform provides its matching service for free. Keeping user attention is therefore central to the platform’s revenue. Advertising-based models remain prominent for different reasons. For example, users prefer to sample a service before committing to payment. A free ad-supported tier is a possible way to attract users who later upgrade to paid subscriptions (Deng et al., 2023). Additionally, privacy concerns provide a further rationale: rather than sharing payment information, many users prefer to trade access to their personal data for a free service, reflecting a money-for-privacy trade-off in digital markets (Kummer and Schulte, 2019). Alternatively, platforms collect search fees from active users, such as pay-per-click or monthly subscription fees. These fees are typically low, distinguishing them from participation fees, which are far less common but used by some selective matching platforms.

The key assumption of the matching rule, Assumption 1, is that each agent receives a recommended match in every period whenever feasible.¹¹ As many online platforms take on a dual role as attention intermediaries and need to attract consumers’ attention to sell to advertisers, providing a constant stream of potential matches aims at grabbing and keeping consumers’ attention.¹² In reality, the attention grabbing component is supported by push notifications or emails preying on the users’ fear of missing out.¹³

4. ANALYSIS

To analyze the equilibrium, I need to characterize the agents’ behavior and the platform’s optimization problem. The agents’ search process is characterized by a set of participation

¹¹In search-and-matching models, time is often modeled as continuous, with matching opportunities arriving at a constant rate. For example, Antler et al. (2023, 2026) make this assumption in the context of a matchmaking platform.

¹²Aridor (2025) shows in a recent experiment that when users face time constraints on a specific platform, they reallocate attention across product categories and to offline activities, suggesting that competition for attention spans multiple markets.

¹³In a recent lawsuit against the MatchGroup Inc. (Oksayan v. MatchGroup Inc., N.D. Cal., No. 3:24-cv-00888, 2/14/24) the plaintiff alleges that the company monopolizes user attention, claiming that “Push Notifications prey on users’ fear of missing out on any potential matches with a strategic notification system designed to capture and retain attention throughout the day”.

and incentive constraints that determine whether an agent is willing to incur the search costs as well as accepts or rejects a recommended match.

4.1 PRELIMINARIES

Consider the strategy of agent θ_i^k being active in the matching market. Upon meeting an agent θ_j^{-k} , the agent decides whether to accept or reject a *recommended* match. Mutual acceptance results in a *match*, and both agents leave the market as a pair. If at least one of the agents rejects the match, agent θ_i^k continues to search. Due to the stationarity of the environment, the continuation value of agent θ_i^k , $V^C(\theta_i^k)$, is defined by the following recursive equation

$$V^C(\theta_i^k) = \delta \omega_i^k + (1 - \delta) \left[-s_i^k + \sum_j \alpha(\theta_i^k, \theta_j^{-k}) \phi(\theta_j^{-k} | \theta_i^k) \theta_i^k \theta_j^{-k} + (1 - \sum_j \alpha(\theta_i^k, \theta_j^{-k}) \phi(\theta_j^{-k} | \theta_i^k)) V^C(\theta_i^k) \right].$$

The first term represents the case in which agent θ_i^k becomes inactive with probability δ and gets its outside option ω_i^k . If the agent remains active with probability $1 - \delta$, it incurs the search cost s_i^k . The expected utility from leaving in a match is given by the utility from a match with type θ_j^{-k} , which is equal to the product of both types, and the probability of meeting and mutually accepting type θ_j^{-k} . With the counterprobability, the match was not mutually accepted and agent θ_i^k continues to search.

$$\sigma_k(\theta_i^k, \theta_j^{-k}) = \begin{cases} 0 & \text{if } \theta_i^k \theta_j^{-k} < V^C(\theta_i^k) \\ r \in [0, 1] & \text{if } \theta_i^k \theta_j^{-k} = V^C(\theta_i^k) \\ 1 & \text{if } \theta_i^k \theta_j^{-k} > V^C(\theta_i^k) \end{cases}, \text{ for } k = A, B. \quad (3)$$

If the match value with a type $\hat{\theta}_j^{-k}$ is larger than the continuation value, agent θ_i^k will accept a recommended match with agent $\hat{\theta}_j^{-k}$ and all agents of types higher than $\hat{\theta}_j^{-k}$. The optimality of this strategy follows directly from the supermodularity of the match payoff. Lastly, an agent participates if the continuation value is larger than the agent's outside option. Due to the stationarity and history-independence of strategies, if an agent decides to participate in the matching market, they will remain active until they are matched or become inactive with probability δ .

Remark. The strategy of an agent of type θ_i^k is increasing in its second argument $\sigma_k(\theta_i^k, \theta_{N-k}^{-k}) \geq \sigma_k(\theta_i^k, \theta_{N-1}^{-k}) \geq \dots \geq \sigma_k(\theta_i^k, \theta_1^{-k})$, but may be neither in- nor decreasing in its first argument.

The fact that the agent's strategy is increasing in its second argument follows directly from Equation 3. If the agent's outside options are weakly increasing in type, for matching rules such as random or positive assortative matching rules $\sigma_k(\theta_i^k, \theta_j^{-k})$ is additionally decreasing in its first argument: $\sigma_k(\theta_{N^k}^k, \theta_j^{-k}) \leq \dots \leq \sigma_k(\theta_1^k, \theta_j^{-k})$. A random matching rule yields the same meeting probabilities for all types. Due to the supermodularity of the payoff function, higher types will reject (weakly) higher types than lower types do. With positive assortative matching, the matching probabilities conditional on being a higher type first-order stochastically dominates the matching probabilities conditional on being a lower type. Hence, higher types will reject strictly higher types than lower types do. In contrast, a negative assortative matching rule, which recommends (almost exclusively) higher types to lower types, and vice versa, can cause lower types to reject lower types while higher types are willing to accept them.

Given the agent's strategy in Equation 3, the acceptance probabilities satisfy

$$\alpha(\theta_i^k, \theta_j^{-k}) = \begin{cases} 0 & \text{if } \theta_i \theta_j < V^C(\theta_i^k) \text{ or } \theta_i \theta_j < V^C(\theta_j^{-k}) \\ 1 & \text{if } \theta_i \theta_j > V^C(\theta_i^k) \text{ and } \theta_i \theta_j > V^C(\theta_j^{-k}) \end{cases} . \quad (4)$$

Equation 4 establishes the relationship between acceptance probabilities and matching outcomes. Mutual acceptance requires that whenever two types of agents meet, both must find it optimal to stop searching.

4.2 LINEAR PROGRAMMING FORMULATION WITH FINITE TYPES

Consider the case with N^k types of agents such that $\Theta^k = \{\theta_1^k, \dots, \theta_{N^k}^k\}$ on side $k = A, B$, where $\theta_{N^k}^k > \dots > \theta_1^k$. The following section provides general results on the existence of an equilibrium, optimal solution and its properties. Let s_i^k be exogenous.

Lemma 1. *For a given feasible matching mechanism, a steady-state equilibrium exists if and only if Equation 2 and 4 are satisfied.*

Suppose for a feasible matching mechanism, an equilibrium exists. Then, it must give rise to (i) a steady state and (ii) optimal strategies of agents, i.e. satisfy Definition 2 and Definition 3. Hence, by (i) Equation 2 (balance condition) must hold, and (ii) implies Equation 4 (optimal mutual acceptance) must hold. Conversely, if Equation 2 is violated the steady state (balance) condition fails and if Equation 4 is violated at least some agent behaves suboptimal. Thus, a feasible matching rule gives rise to an equilibrium if and only if Equation 2 and 4 hold.

Lemma 2. *There exists a feasible matching rule that gives rise to an equilibrium.*

In the most simple case consider the matching rule $\phi(\omega_i^k | \theta_i^k) = 1$ for all types $\theta_i^k \in \Theta^k$ on side $k = A, B$. Given that agents are matched with their outside option, no agent

is willing to incur search costs. With no agent participating in the steady state, the matching rule is feasible and gives rise to a steady state equilibrium.

Next, to determine the profit-maximizing matching rule $\mathcal{M}$, it is useful to define the matching outcome. Intuitively, the matching outcome is defined as the matrix that describes the distribution of pairs under matching rule $\mathcal{M}$ that exit in a match. Recall that matrix M describes the masses of recommended pairs under matching rule $\mathcal{M}$ and let A denote the matrix of agents' mutual acceptance probabilities

$$A \equiv \begin{pmatrix} \alpha(\theta_1^k, \theta_1^{-k}) & \cdots & \alpha(\theta_1^k, \theta_{N-k}^{-k}) \\ \vdots & & \vdots \\ \alpha(\theta_{N^k}^k, \theta_1^{-k}) & \cdots & \alpha(\theta_{N^k}^k, \theta_{N-k}^{-k}) \end{pmatrix}.$$

Formally, the matching outcome is defined as the componentwise multiplication (Hadamard product) of matrix A and M :

Definition 4. The matching outcome is defined by the matrix

$$A \odot M = \begin{bmatrix} \alpha(\theta_1^k, \theta_1^{-k})\Phi(\theta_1^k, \theta_1^{-k}) & \cdots & \alpha(\theta_1^k, \theta_{N-k}^{-k})\Phi(\theta_1^k, \theta_{N-k}^{-k}) \\ & \vdots & \\ \alpha(\theta_{N^k}^k, \theta_1^{-k})\Phi(\theta_{N^k}^k, \theta_1^{-k}) & \cdots & \alpha(\theta_{N^k}^k, \theta_{N-k}^{-k})\Phi(\theta_{N^k}^k, \theta_{N-k}^{-k}) \end{bmatrix} \equiv O(\cdot).$$

Matching outcomes are (i) assortative if $O(\cdot)$ has positive entries only along the main diagonal, (ii) weakly assortative if $O(\cdot)$ has positive entries along the main diagonal and to the right if and only if all entries below are also positive, and (iii) non-assortative otherwise.

If a matching outcome is assortative, this implies that lower types are matched with strictly lower types than higher types while the definition of weakly assortative implies that lower types can be matched with the same types as higher types. The definition is weak in the sense that it does not require that lower types accept with a higher probability than higher types. Other matching outcomes are called non-assortative and entail negative assortative outcomes where higher types are matched with strictly lower types than lower types.

Denote by $m(\theta_i^k, \theta_j^{-k}) = \alpha(\theta_i^k, \theta_j^{-k})\Phi(\theta_i^k, \theta_j^{-k})$ an entry of matrix $O(\mathcal{M})$. Each entry is therefore the mass of *matched* pairs that exit the market together in every period. For a given matching rule, an equilibrium induces at most one matching outcome since the mutual acceptance probabilities and steady state masses are pinned down in equilibrium.

To find the profit-maximizing matching rule and the associated matching outcome, I proceed in two steps. First, I fix a matrix of acceptance probabilities and determine the optimal feasible matching rule that implements the mutual acceptance probabilities.

Second, supposing the optimal matching rule from step one is used to implement any chosen matrix of acceptance probabilities, I choose the matrix that yields the highest platform profits. Note that the platform finds it optimal to induce either full participation of a type or no participation.

Lemma 3. *It is without loss of generality to consider $\eta_i^k \in \{0, 1\}$.*

Suppose the platform charges type-dependent search fees, and type θ_i^k , who is indifferent between participating and not participating, participates with probability less than one. Then, the platform makes the same profit if type θ_i^k participates with probability one and the platform sometimes matches them to their outside option while reducing their search fee to get the same payments in expectation. If the platform uses an advertising-based business model, the platform will strictly increase its profit by this procedure due to the concavity of advertising. Therefore, from now on I will focus on $\eta_i^k \in \{0, 1\}$, which allows to focus on the set of participating types. Then, suppose the platform induces a set $\hat{\Theta}^k$ for $k = A, B$ to participate.

In the following, I will transform the platform's profit-maximization problem into a linear program. For given search cost s_i^k , recall that the platform's objective is to maximize

$$\max_{\mathcal{M}} \sum_{k=A,B} \sum_{\theta_i^k \in \hat{\Theta}^k} \frac{(1-\delta)\nu(s_i^k)}{1-\rho} f(\theta_i^k),$$

i.e., the platform maximizes the steady state mass of active agents with weight $\nu(s_i^k)$. Note that the platform does not earn revenue from agents that are inactive or do not participate in the market in the first place. The maximization problem underlies a set of constraints. First, the matching rule must implement a steady state. The steady state condition (Equation 2) implies that the inflow of agents θ_i^k is equal to the mass of agents that become inactive in a period with probability δ and the mass of active agents that exit in matched pairs. In the steady state, the mass of agents of type θ_i^k can be restated as

$$f(\theta_i^k) = \frac{1}{\delta} \left(\beta_i^k - (1-\delta) \sum_j m(\theta_i^k, \theta_j^{-k}) \right), \quad (\text{Steady-State Mass})$$

and therefore depends positively on the inflow, β_i^k , and negatively on the mass of matched pairs that include type θ_i^k . Second, the matching rule determines whether agents participate in the market and whether agents search according to the platform's recommendations. For participating agents, it must hold that the agent prefers participating in the

market to accepting the outside option, i.e.

$$\omega_i^k \leq \frac{\delta \omega_i^k + (1 - \delta) \left(-s_i^k + \sum_j \alpha(\theta_i^k, \theta_j^{-k}) \phi(\theta_j^{-k} | \theta_i^k) \theta_i^k \theta_j^{-k} \right)}{\delta + (1 - \delta) \left(\sum_j \alpha(\theta_i^k, \theta_j^{-k}) \phi(\theta_j^{-k} | \theta_i^k) \right)} = V^C(\theta_i^k).$$

Since the match payoffs are supermodular, there exists a critical lowest type that an agent θ_i^k is willing to accept (Equation 3). Agent θ_i^k rejects (accepts) all types below (above) the critical lowest type. The incentive constraint for agent θ_i^k to follow the recommendation of the platform to (weakly) reject an agent θ_j^{-k} reads¹⁴

$$\theta_i^k \theta_j^{-k} \leq V^C(\theta_i^k).$$

By using the steady state condition, the participation and incentive constraints can be reformulated. Note that the denominator of the continuation value is equal to the probability that an agent exists, which is equal to $\beta_i^k / f(\theta_i^k)$ by Equation 2. Inserting into the continuation value and rearranging yields

$$\beta_i^k \omega_i^k \leq \delta f(\theta_i^k) \omega_i^k - (1 - \delta) f(\theta_i^k) s_i^k + (1 - \delta) \sum_j \alpha(\theta_i^k, \theta_j^{-k}) \Phi(\theta_j^{-k} | \theta_i^k) \theta_i^k \theta_j^{-k}, \quad (\text{PC})$$

$$\beta_i^k \theta_i^k \theta_j^{-k} \leq \delta f(\theta_i^k) \omega_i^k - (1 - \delta) f(\theta_i^k) s_i^k + (1 - \delta) \sum_j \alpha(\theta_i^k, \theta_j^{-k}) \Phi(\theta_j^{-k} | \theta_i^k) \theta_i^k \theta_j^{-k}, \quad (\text{IC})$$

where $\alpha(\theta_i^k, \theta_j^{-k}) \Phi(\theta_j^{-k} | \theta_i^k) = m(\theta_i^k, \theta_j^{-k})$. Lastly, the platform's matching rule must satisfy the feasibility constraints. Without loss of generality, let side B be of smaller or same size as side A . Then on side A , the sum over the mass of each recommended pair that includes type θ_i^A must be equal to the steady state mass of θ_i^A . On side B , the sum over the mass of each recommended pair that includes type θ_i^B and the mass of agents of type θ_i^B that are unmatched must be equal to the steady state mass of type θ_i^B

$$\sum_{\theta_j^{-k} \in \Theta^{-k}} \Phi(\theta_i^k, \theta_j^{-k}) + \mathbf{1}_{k=A} \Phi(\theta_i^k, \omega_i^k) = f(\theta_i^k), k = A, B. \quad (\text{Feasibility})$$

As stated above, for given matrix A the above constraints and the objective function are all linear functions of the mass of matched and recommended pairs $\{m(\theta_i^k, \theta_j^{-k}), \Phi(\theta_i^k, \theta_j^{-k})\}_{ij}$. The steady-state masses $f(\theta_i^k)$ are affine functions of the mass of matched pairs by Equation (Steady-State Mass). To complete the reformulation as linear program, it remains to include the indifference constraints for agents who mix when accepting type from the other market side, which implies that the respective incentive constraint must hold with equality. Appendix A.1 formally does so, leading to:

¹⁴In mechanism design, this is often referred to as an obedience constraint because there is no private information throughout the model.

Lemma 4. *The platform's problem can be restated as a linear programming problem in the mass of matched and recommended pairs: $\{m(\theta_i^k, \theta_j^{-k})\}, \{\Phi(\theta_i^k, \theta_j^{-k})\}_{ij}$.*

Note that by Lemma 1, the solution to the linear program is an equilibrium as it fulfills Equation 2 and 4. Given a solution of the linear program, the optimal matching rule to the original problem results from

$$\phi(\theta_j^{-k} | \theta_i^k) = \frac{\Phi(\theta_i^k, \theta_j^{-k})}{f(\theta_i^k)}.$$

Next, I show that the auxiliary problem has an optimal solution. I say that a matrix A of mutual acceptance probabilities can be implemented if there exists a matching mechanism $\mathcal{M}$ such that $((f(\theta_i^k))_{\theta_i^k \in \Theta^k}, A, \eta)$ is an equilibrium. Let $\mathcal{A}$ be the set of matrices A that can be implemented. By Proposition 2, $\mathcal{A}$ is non-empty. For every $A' \in \mathcal{A}$, construct a matrix A'' such that

$$\begin{aligned} \alpha''(\theta_i^k, \theta_j^{-k}) &= \alpha'(\theta_i^k, \theta_j^{-k}) \text{ if } \alpha'(\theta_i^k, \theta_j^{-k}) \in \{0, 1\}, \\ \alpha''(\theta_i^k, \theta_j^{-k}) &= \alpha_{ij} \text{ otherwise,} \end{aligned}$$

where α_{ij} can take on any value in $[0, 1]$. I use $\alpha_{ij} \in [0, 1]$ whenever an agent is indifferent, which implies that the same constraints in the auxiliary program must hold. Denote the resulting set of matrices as $\mathcal{A}^*$ and note that $\mathcal{A}^*$ is finite. Now, I can solve the linear program over the mass of matched and recommended pairs (ignoring acceptance probabilities). Solving this for all (finite) possible combinations of constraints yields a set of candidate solutions among which I choose the one that maximizes the platform's profit. To find the corresponding acceptance probabilities $\alpha_{ij} \in [0, 1]$ when the agent is indifferent, divide the matched pairs through the recommended ones

$$\alpha_{ij} = \frac{m(\theta_i^k, \theta_j^{-k})}{\Phi(\theta_i^k, \theta_j^{-k})}.$$

Formally, as $\mathcal{A}^*$ is finite, only a finite number of linear problems must be solved. Each linear program returns a set of candidate solutions and a value of the objective function. Fixing $A \in \mathcal{A}^*$, the linear program returns a value $\Pi(A)$, i.e., the profit level, and let $\mathcal{G} = \bigcup_{A \in \mathcal{A}^*} \Pi(A)$ be the set of profit levels for all linear programs with $A \in \mathcal{A}^*$ that implement an equilibrium.

Lemma 5. *The set $\mathcal{G}$ is non-empty and finite with $\Pi(A) < \infty$ for all $A \in \mathcal{A}^*$ and $-\infty < \Pi(A)$ for at least one $A \in \mathcal{A}^*$.*

Key to the proof is to show that the linear program for any given matrix $A \in \mathcal{A}$ is (i) not unbounded and (ii) not infeasible, i.e. the feasible region is non-empty. Given that

both (i) and (ii) are satisfied, an optimal solution to the linear program exists and the linear program attains a finite optimal value (Dantzig, 1963).¹⁵

Theorem 1. *There exists an optimal solution.*

I proceed by showing that an optimal solution exists for any exogenous search costs s_i^k for all $\theta_i^k \in \Theta^k$, $k = A, B$. By Lemma 5, the maximum over set $\mathcal{G}$ is well-defined as $\mathcal{G}$ is finite and bounded such that an optimal solution exists. Next, I show that there exists an optimal solution if the platform chooses search costs s_i^k for all $\theta_i^k \in \Theta^k$, $k = A, B$. Through a series of Lemmas, I prove that the set $\mathcal{G}$ is compact-valued and upper hemicontinuous in the vector of search costs. This implies that the set $\max \mathcal{G}$ is upper semicontinuous in the vector of search costs. Therefore, by an extension of the Weierstrass theorem a maximum exists.

4.3 PROPERTIES OF THE OPTIMAL SOLUTION

To identify properties of the optimal solution, first consider two prominently studied matching rules. As discussed in Section 2, in decentralized matching-and-search markets agents are often assumed to meet according to a random matching technology. A natural question to consider is whether a platform that has access to extensive user data would commit to a random meeting technology as well.

Proposition 1. *Suppose $N^k \cdot N^{-k} > 1$. Random matching is generically suboptimal for exogenous search costs as well as endogenous search fees. Consider the class of functions: $\nu(s_i^k) = \kappa(s_i^k)^\alpha$ with $\kappa \in \mathbb{R}^+$ and $\alpha \in (0, 1)$. Random matching is generically suboptimal within this class of functions.*

The proposition shows that random matching is generically suboptimal for the platform if search costs are exogenous or type-dependent search fees are endogenously chosen.¹⁶ For analytical convenience, I consider the class of concave revenue functions in the proof to determine a knife-edge solution. Consider the nontrivial case in which there are different types to be matched. Under random matching, the conditional probability of meeting a type θ_i^k on side k is the same for all types $\theta_j^{-k} \in \Theta^{-k}$ on side $-k$ and corresponds to the proportion of type θ_i^k in the population. As shown in Appendix C.1, the probability of meeting a type θ_i^k is a function of the inflow, β_i^k , and the probability

¹⁵Existence follows from the fact that the constraint set is a convex polyhedron. Because the objective is linear and the constraint set is convex, any local extremum will be the global extremum. As the objective is linear, the extremum will be obtained at one of the extreme points of the constraint set, i.e., at the vertices of the polyhedron.

¹⁶Consider the following definition for *generically suboptimal*. The probability of the case in which random matching is optimal occurs with probability zero when the model parameters are randomly drawn from continuous intervals as defined in the proof.

of exit, δ . In contrast, for given search costs, the optimal solution of the linear program is a function of these and internalizes changes in the search cost. Therefore, random matching is generically suboptimal for given search costs, although it may coincide with the optimal solution for knife-edge $s_i^k, \theta_i^k, \delta$, and β_i^k . Consider next the case in which the platform chooses type-dependent search fees. The platform does not choose random matching but instead implements the positive assortative matching rule that maximizes the agents match surplus and extracts all surplus via the search fee.

Proposition 1 highlights that a platform, which has increasing access to user data, generically does not commit to a random matching technology. The result implies that data is valuable because it allows the platform to condition meeting probabilities on user types and thereby improve upon type-independent matching

Corollary 1. *Suppose a platform has access to data about user types. Generically, the platform makes higher profits by using the data to discriminate users by conditioning the matching rule on user types instead of refraining from using user data.*

Second, consider the positive assortative matching rule (PAM) under the assumption that both sides are symmetric with respect to the inflow of new agents: $\beta_i^A = \beta_i^B$, their type space $\Theta^k = \Theta$, and outside options. Under symmetry, PAM matches agents if and only if they are of the same type on both sides of the market. In this particular case, PAM is of special interest in the literature as it maximizes total match surplus when the match utility is supermodular, where an agent's individual match surplus is defined as the difference between the expected match utility on the platform and the agent's outside option. Furthermore, the resulting positive assortative matching outcome, is equivalent to the set of stable matchings (Roth and Sotomayor, 1992). That is, matches are individually rational, i.e., yield a utility greater than their outside option, and are pairwise stable, i.e., there exists no blocking pair of agents that would prefer to be matched to each other instead of the equilibrium matching. The next proposition shows under which circumstances the positive assortative matching rule (PAM) is not profit-maximizing.

Proposition 2. *Suppose both market sides are symmetric.*

- (i) **(Price Ceiling)** *PAM is profit-maximizing if the platform can charge arbitrary high type-dependent search fees. Conversely, for every type $\theta_i \in \Theta \setminus \{\theta_1\}$ there exists a threshold $\bar{s}_i$ such that if $s_i < \bar{s}_i$, PAM is suboptimal.*
- (ii) **(Advertising)** *There exists a threshold $\bar{\delta}$ such that if $\delta \leq \bar{\delta}$ and $\nu(\cdot)$ is concave, PAM is suboptimal.*
- (iii) **(Complexity-Constrained Pricing)** *If the platform can only set $N - 1$ search fees but finds it optimal to serve all N types, PAM is suboptimal.*

Proposition 2 raises the question of why users do not observe high search fees online, and why matching appears to be worsening rather than improving. The analysis highlights a central trade-off. Positive assortative matching maximizes match surplus and is optimal when the platform can fully extract the surplus through high search fees. When surplus extraction is constrained, however, the platform optimally distorts matching away from PAM by inducing mismatches that prolong search and increase revenue.

Limited Rent Extraction: Price Ceiling. First, PAM is indeed profit-maximizing if the platform has pricing power. When the platform commits to a (time-constant) deterministic matching rule such as PAM, agents will accept the recommended match in the first period. Therefore, all agents search for exactly one period, which results in a steady state population equal to the inflow for each type. By charging (high) type-dependent fees, the platform can extract the full surplus from agents, i.e., the expected match value of an assortative match over the agent’s outside option. In this case, the “search fee” is paid once, since agents search for only one period.

The proposition, however, shows that if the platform cannot commit to high search fees, for example due to a (binding) price ceiling $\bar{s}$, then PAM is no longer optimal. Let $\bar{s}$ be such that s_i violates the condition in Proposition 2 for at least one type $\theta_i \in \Theta \setminus \{\theta_1\}$. Then the platform can no longer extract the full surplus from an agent of type θ_i . Then, PAM is not profit-maximizing, as the platform has an incentive to deviate to a matching rule under which type θ_i and the lowest type θ_1 meet with mass ε and additionally decrease the fee for the lowest type s_1 . The price ceiling $\bar{s}$ is such that whenever type θ_i and type θ_1 meet, θ_i (weakly) rejects θ_1 under the new matching rule. This implies that type θ_i searches more than one period, increasing the payments collected from θ_i . Thus, given the pricing constraint the platform cannot extract the full surplus and substitutes surplus extraction via the price with surplus extraction via prolonged search times.

As discussed in Section 3.1, users are often reluctant to commit to large upfront payments, which limits the platform’s pricing power. Additionally, fees for in-app purchases for instance in Apple’s App store are capped at \$999.99, whereas the estimated lifetime utility from a match and hence, potential willingness to pay for a partner could be well above \$999.99.¹⁷

Advertising. Second, suppose the platform follows an advertising-based business model. If the return to advertising is concave and $\delta \leq \bar{\delta}$, then PAM is suboptimal. Under PAM agents search for only one period. Thus, a profit-maximizing platform would need to

¹⁷Indeed, traditional matchmakers charge over ten times the amount; see https://www.nytimes.com/2024/02/13/business/dating-bounty-roy-zaslavskiy.html?unlocked_article_code=1.VU0.XqAb.q2iJT-p0bHz1&smid=nytcare-ios-share&referringSource=articleShare

impose the highest feasible search cost per agent. With concave advertising returns, however, it becomes more profitable to reduce search costs and increase the mass of participating agents. In other words, users switch their attention if the platform advertises too much, as is well known from other contexts (Anderson and Coate, 2005). Thus, doubling the amount of advertisement does not double the revenue. Since $\delta > 0$ implies a loss in profits due to exogenous attrition that increases with longer search times, a high δ reduces the platform’s willingness to trade off longer search durations for lower costs.

An advertisement-based business model generates higher profits than charging personalized prices if advertisement revenue is sufficiently efficient compared to its nuisance. For general revenue functions $\nu(s_i)$, a sufficient condition is

$$\frac{\nu(s_i)}{s_i} \geq \frac{\theta_i^2}{s_i \cdot \tau(\theta_i)}, \forall \theta_i \in \Theta,$$

where the numerator is the full surplus that can be extracted from agents under PAM with personalized fees and the denominator is the total amount of search cost collected during agents’ search with advertisement, with $\tau(\theta_i)$ being the time spent on the platform.

Advertisement plays a key role in the digital economy.¹⁸ More specifically, in the light of the application to matching platforms, a substantial share of these platforms rely on advertisement as a source of revenue. As discussed in Section 3, advertising monetizes attention without requiring payment systems, making it attractive when users are reluctant to pay directly. Additionally, the model can allow for targeted advertising, i.e., more data about the user might imply that the platform can extract higher advertising rent from each type. As the results (Theorem 1 and Proposition 1 and 2) do not hinge on $\nu(\cdot)$ being type-independent, the model could substitute $\nu(s_i^k) = \nu_i^k(s_i^k)$ for type-dependent revenue. The results do not change qualitatively.

Complexity-Constrained Pricing: Third, suppose that two out of N types face the same search fee. If the platform finds it optimal to exclude one of the two agents, e.g., by setting a price higher than their willingness to pay for the service, then the platform can extract the match surplus from the other type by implementing PAM. If, however, it is suboptimal to exclude one type, then depending on the inflow of types, the platform maximizes profit by using a matching rule different from PAM. In other words, a uniform fee induces a pricing distortion which is compensated by a matching distortion: since the platform cannot price-discriminate between types facing the same fee, it instead distorts the matching rule to induce type-specific search durations, thereby screening agents

¹⁸Recent evidence shows that policies restricting targeted advertising lead to substantial revenue declines for firms dependent on ad-based business models, highlighting the economic role of advertising revenue (Aridor et al., 2025).

through search duration rather than price. This mechanism is characterized formally in the proof of Proposition 2(iii) and illustrated in the next Section 4.4.

A reason why the platform is constrained in setting agents' search costs is that in reality, a platform serves many types of users, which would require complex pricing schemes to extract each agent's surplus. I therefore consider a setting in which two types of agents face the same search cost designed by the platform. One possible interpretation is that agents choose among a small number of (discrete) pricing tiers, with all agents on the same tier paying the same amount as common on dating platforms. For example, many job platforms charge a uniform price per click within a given job category.

4.4 APPLICATION TO BINARY TYPES

The binary type case delivers a full characterization of the profit-maximizing matching rule and illustrates the two inefficiencies, delayed matching and mismatch. Suppose for ease of exposition that market sides are symmetric. There are two types on each side of the market with a strictly positive inflow. With slight abuse of notation, denote the type set by $\Theta = \{\theta_h, \theta_l\}$ with $\theta_h > \theta_l$. Outside options are $\omega_l < \theta_l^2$ and $\omega_h < \theta_h \theta_l$ to exclude environments in which acceptance behavior of agents is driven by the outside option rather than the matching rule.

With two types, the mutual acceptance matrix takes the following form

$$A = \begin{bmatrix} \alpha(\theta_h, \theta_h) & \alpha(\theta_h, \theta_l) \\ \alpha(\theta_l, \theta_h) & \alpha(\theta_l, \theta_l) \end{bmatrix},$$

where the mutual acceptance probability of the assortative matches are along the diagonal and the mutual acceptance probability of mismatches are off the diagonal. In symmetric equilibria, the solution satisfies $\alpha(\theta_h, \theta_l) = \alpha(\theta_l, \theta_h)$. Then with two types, only three possible matrices can be implemented as part of an equilibrium

$$A_{PAM} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, A_{WPAM} = \begin{bmatrix} 1 & \alpha'_{ij} \\ \alpha'_{ij} & 1 \end{bmatrix}, A_{NAM} = \begin{bmatrix} 1 & \alpha''_{ij} \\ \alpha''_{ij} & 0 \end{bmatrix}, \alpha'_{ij}, \alpha''_{ij} \in [0, 1].$$

Given the platform's matching rule, high types either accept only high types or accept low types with positive probability resulting in three possible matrices. If high types accept only high types, low types accept high and low types, resulting in positive assortative acceptance (A_{PAM}). If high types accept low types with positive probability, low types may also accept low types, resulting in weakly assortative acceptance (A_{WPAM}). Alternatively, low types may reject low types, resulting in a non-assortative acceptance in which high types accept low types, but low types do not (A_{NAM}).

For each of the three possible acceptance matrices, there exists an optimal matching

rule that can implement the corresponding outcome over a range of parameter values (Theorem 1). Given the existence of an optimal matching rule, which matrix A maximizes the platform's profit for *fixed* search costs?

First, consider the maximum rent that the platform can extract when the positive assortative matrix, A_{PAM} , is implemented. A high type agent is willing to search the longest for a match with another high type. In this case, the maximum rent the platform can extract from a high type agent is proportional to $\theta_h(\theta_h - \theta_l)$, which is the high type's value times the *match premium*. The match premium is the gain from matching with a high type rather than a low type. If the platform were to extract more rent, high types would start accepting low types as well, and thus only search for one period. In turn, if high types always reject low types, the maximum rent the platform can extract from low types is proportional to $\theta_l^2 - \omega_l$.

Due to feasibility constraints, the platform is constrained by the ratio of high to low types when choosing the matching rule. The platform can extract the rent from both types as described above if both the high-type incentive constraint and the low-type participation constraint bind requiring the inflow ratio to satisfy:

$$\left(\frac{\beta_h}{\beta_l}\right)^{(a)} = \frac{(1 - \delta)(\theta_l^2 - s_l)(s_h + \delta(\theta_h^2 - s_h))}{(\theta_h(\theta_h - \theta_l) - s_h - \delta(\theta_h^2 - s_h))(s_l + \delta(\theta_l^2 - s_l))}. \quad (5)$$

At this ratio, high types are just indifferent between accepting and rejecting low types, while low types are just indifferent between participating or not, which results in

$$\phi(\theta_h|\theta_h) = \frac{s_h + \delta(\theta_h\theta_l - s_h - \omega_h)}{(1 - \delta)\theta_h(\theta_h - \theta_l)}, \quad \phi(\theta_l|\theta_l) = \frac{s_l}{\theta_l^2 - \omega_l}.$$

Due to feasibility constraints, however, other than in this knife-edge case the incentive and participation constraints cannot bind at the same time while implementing A_{PAM} . As the ratio increases, more high types enter compared to low types. In this case, the platform makes the participation constraint binding for low types. The probability of a high type meeting a high type must increase such that high types are left with a rent greater than $\theta_h\theta_l$. As the ratio decreases, fewer high types enter compared to low types. The platform makes the incentive constraint binding for high types, leaving a positive rent for low types as the probability of a low type meeting a low type must increase. In both cases, the platform potentially forgoes a significant amount of rent when moving away from the “optimal” ratio.

Second, consider the maximum rent that the platform can extract when A_{WPAM} is implemented. Suppose the ratio of high to low types is greater than in Equation 5. Then, the platform can commit to a matching rule in which high types randomize over accepting and rejecting low types, while low types remain indifferent between participating and their

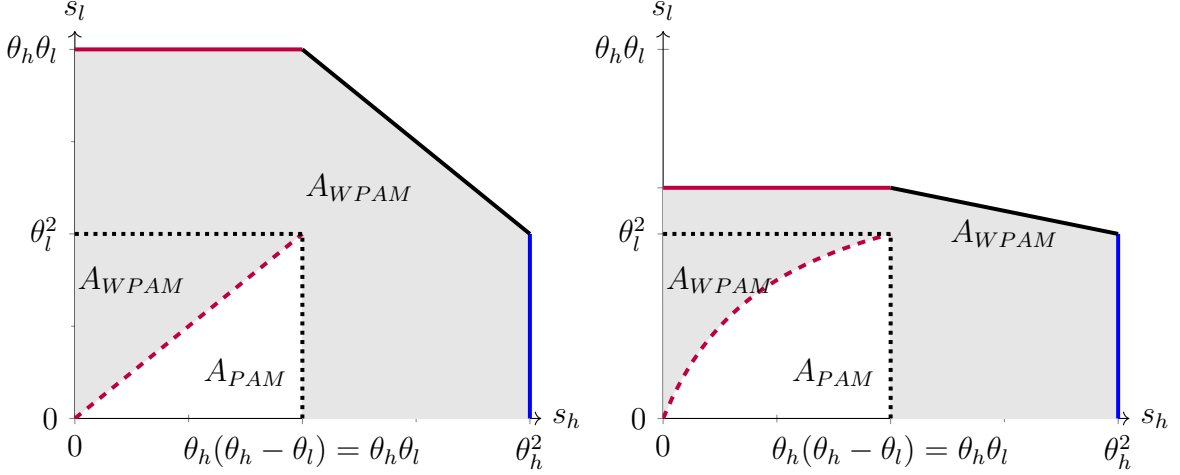


Figure 1: Profit-Maximizing Mutual Acceptance for a 1:1 Inflow (Left) and 1:3 Inflow (Right)

outside option. The expected match utility of high types decreases, while the expected match utility of low types increases. For a ratio of high to low types greater than in Equation 5, implementing A_{WPAM} yields a higher profit than A_{PAM} . When implementing A_{PAM} , the platform must increase the meeting probability of assortative pairs as the ratio β_h/β_l increases, otherwise low types will no longer be willing to participate. This implies, however, that the platform forgoes rent from high types. Inducing high types to accept mismatches with positive probability, $\alpha(\theta_h, \theta_l) > 0$, in contrast leads to a longer search of low types as they receive a higher expected match utility. Increasing $\alpha(\theta_h, \theta_l)$ raises the continuation value of low types, increases their steady-state mass, and thereby relaxes feasibility constraints for high types.

Third, consider the maximum rent that the platform can extract when A_{NAM} is implemented. High types accept both types with positive probability, while low types reject low types and only exit in (mis-)matches with high types. For any search cost, implementing A_{NAM} is dominated by A_{WPAM} . Under A_{NAM} , low types cannot be held at their participation constraint, implying strictly positive rents relative to A_{WPAM} (at least θ_l^2 compared to ω_l).

Figure 1 illustrates the mutual acceptance matrices over the search cost space (s_h, s_l) that maximize the platform's profit taking $\delta \rightarrow 0$.¹⁹ A_{PAM} can only be implemented in the rectangle given by the dotted black line. The dashed red line represents the condition in Equation 5: along this line, A_{PAM} is implemented and both incentive and participation constraints bind simultaneously. Above the line, A_{PAM} can be implemented

¹⁹The profit functions used to build the figure can be found in Appendix B in the proof for Proposition 2 part (iii). The figure uses $\theta_h = 2\theta_l$, $\omega_h = \omega_l = 0$, and $\beta_h : \beta_l = 1 : 1$ (left) or $\beta_h : \beta_l = 1 : 3$ (right) and takes $\delta \rightarrow 0$.

with a binding participation constraint, but the incentive constraint is slack, so A_{WPAM} generates strictly higher profit. Intuitively, A_{WPAM} allows for an additional degree of freedom as the platform chooses $\alpha(\theta_h, \theta_l)$ as well. This allows the platform to increase rent from high types while keeping the participation constraint binding for low. For any $s_h > \theta_h(\theta_h - \theta_l)$, the mutual acceptance must be A_{WPAM} , since the fee exceeds the match premium and a high type would always accept a low type rather than search for another period. Similarly, for any $s_l > \theta_l^2$, the mutual acceptance must be A_{WPAM} , as low types would no longer participate under positive assortative acceptance.

As already shown by Proposition 2, the matching rule PAM is only implemented for high enough search costs, specifically along the thick blue line in Figure 1, which induces A_{WPAM} with matching outcome $\mathcal{O}_{PAM}$. For all other search costs, the platform strategically lowers the quality of recommended matches to induce additional search. The platform's optimal matching rule creates two distinct economic inefficiencies. Within the dashed rectangle, all matching outcomes feature *delay*: the platform fosters mismatches by recommending low types to high types, who then reject them with positive probability, inducing at least one additional period of search. Delaying matches increases the per-period payments the platform collects. In the gray region, the platform fosters *mismatch*: it induces users to leave the market in inefficient pairs rather than waiting for their assortative match. All matching outcomes in the gray region feature mismatches, apart from the knife-edge case of PAM along the blue line. Along the thick red line, the platform implements NAM representing the extreme case in which mismatch is maximized (in the left figure, all high types are matched to low types with probability one). This decomposition illustrates the central mechanism of the model: the platform trades off match quality against search duration, and the relative weight it places on each distortion depends on the search cost parameters and the inflow ratio.

The qualitative structure of Figure 1 is unchanged under advertising. Advertising does not alter the set of implementable matching rules or outcomes, but changes the profitability associated with each point in the (s_h, s_l) -plane and thereby the optimal choice of search costs. Part (i) of Proposition 2 restricts the feasible set of search costs, effectively truncating the region in which matching rule PAM is implemented. Part (iii) imposes a linear pricing constraint, corresponding to the 45-degree line. With a balanced inflow, this implies that A_{WPAM} is implemented throughout the feasible search cost space. With an unbalanced inflow, the constraint intersects the A_{PAM} region and subsequently crosses the boundary characterized by Equation 5, beyond which A_{WPAM} becomes optimal.

5. DISCUSSION

Competition Although digital markets are often characterized by substantial market power held by a few large platforms, matching platforms also face competition from a fringe of smaller providers. To capture these competitive forces, introduce a competitive fringe while fixing any given business model, so that agents continue to pay (implicitly) per period. Agents single-home, joining only one platform, and competitors' strategies are taken as given. Competition then increases agents' outside option, i.e., the elasticity of participation.

The core structure of the model is unaffected by this extension. The linear programming formulation, and thus the existence of an optimal solution (Theorem 1), is independent of the outside option. Likewise, the result that random matching is generically suboptimal (Proposition 1) continues to hold. The argument extends by considering outside options drawn from an interval induced by competition. The first part of Proposition 2 is also unchanged: PAM remains profit-maximizing when the platform can charge arbitrarily high search fees. In contrast, the conditions characterizing the other cases are affected. Under a price ceiling, an increase in the outside option lowers the maximal price that would induce the platform to no longer implement PAM. Under uniform pricing, the qualitative insights remain similar, though matching outcomes tend to become more assortative. Under advertising, the effect of competition depends on the inflow of agents, which determines whether the relevant threshold in δ shifts upward or downward.

Network Effects and Non-Stationarity There are multiple sources that could give rise to non-stationarity in the model. Structurally, the model is a once-and-for-all matching market: once a match forms, both agents exit permanently. As emphasized by Shimer and Smith (2001), non-stationarity arises in environments where matches can be broken; a mechanism that is not available here, as the platform cannot dissolve matches. Allowing for non-stationarity by conditioning on users' time on the platform, would require solving a sequence of complex assignment problems. Under type-dependent search fees and positive assortative matching, however, agents exit immediately so that non-stationarity is irrelevant for the optimal solution of the platform. More generally, building up a user base to exploit it in later periods would require leaving rents to agents along the transition path, limiting the scope for dynamic manipulation.

Network effects in the model are endogenous and operate through meeting probabilities. When market sides are asymmetric, an increase in the smaller side raises meeting rates for agents on the larger side. Users, however, derive utility only from match quality, not from the number of users per se, and match quality is itself endogenously determined by the matching rule. Moving away from the steady state could, in principle, make delay

desirable if the platform benefits from improving the composition of the market over time. Such dynamic network effects are not captured here. The analysis therefore focuses on stationary environments in which the platform optimally distorts matching within the steady state rather than through intertemporal manipulation of the user base.

6. CONCLUSION

On matching platforms, the misalignment of incentives between users and the platform becomes more problematic as platforms possess more data and develop more predictive algorithms. This paper presents a model in which a platform has perfect information about its users' types and matches them to its advantage. I show how the platform utilizes user information to improve upon random matching — the case where the platform has no information about its users' types. Moreover, I characterize conditions under which the platform deviates from the socially optimal matching rule (PAM). Under these conditions, the platform's matching creates two intertwined inefficiencies: it distorts matching outcomes by inducing mismatches that deviate from the socially optimal outcome, and it increases users' search time, leading to higher search costs than necessary.

The results imply that policymakers aiming to reduce mismatch and search duration in matching markets, should be concerned about matching platforms. Rather than relying on platforms to reduce search frictions, the platform's algorithm is itself a source of mismatch and welfare loss. This highlights the importance of algorithm audits that have been discussed in light of the EU Digital Services Act (Panigutti et al., 2025).²⁰ After uncovering the potential risk of matching algorithms theoretically, future research or audits could be risk-measuring studies as proposed by Panigutti et al. (2025) and employ experimental or observational methods to empirically evaluate the harm created by matching algorithms.

The platform's incentive to prolong search and induce mismatch has direct implications for real-world markets. In labor markets, mismatch reduces productivity and contributes to long-term unemployment (Şahin et al., 2014; McGowan and Andrews, 2015), while in marriage markets, assortative mating shapes household inequality (Pestel, 2017; Eika et al., 2019; Almar et al., 2023). Whether existing platforms generate these distortions in practice remains an open empirical question. In dating markets, Hitsch et al. (2010) show that matches are approximately efficient and stable. The authors, however, rely on data before the advent of large, algorithmic dating platforms. In contrast,

²⁰Similar initiatives have been discussed by the Digital Regulation Cooperation Forum in the UK; see <https://www.gov.uk/government/publications/findings-from-the-drcf-algorithmic-processing-workstream-spring-2022/auditing-algorithms-the-existing-landscape-role-of-regulators-and-future-outlook>

more recent evidence, such as Sharabi and Dorrance-Hall (2024), finds that people who meet online are less satisfied in their marriages. In labor markets, Kroft and Pope (2014) show that Craigslist has no effect on the unemployment rate. Similarly, Gürtzgen et al. (2021) provide evidence that online searches do not affect employment stability or wage outcomes, but instead increase the proportion of unsuitable candidates in job applications. Taken together, this evidence is consistent with the paper’s prediction that platform intermediation need not improve matching outcomes and may worsen them.

Methodologically, the paper introduces a linear programming reformulation of dynamic matching problems with endogenous meeting rules. The reformulation can be extended to other applications whenever the designer’s payoff is linear in the mass of users and users search for matches. The reformulation does not depend on the agent’s payoff or production function and thus, allows for a broader class of models. For example, the linear program can be applied to various (dynamic) platform design problems such as online marketplaces with ranking or recommendation algorithms. Fixing search costs and the mutual acceptance matrices, a ranking and recommendation algorithm can be understood as facilitating matches and search between buyers and sellers. The model can be simplified by modeling one market side as non-strategic, for example as products, hotels, or content. Adjusting the mutual acceptance matrix and altering the set of necessary constraints, the LP can be applied again.

Lastly, the analysis focuses on a monopoly platform in a stationary environment and abstracts from dynamic network effects and non-stationarity. Future research could extend the model along these dimensions, for example analyze competition and entry in matching markets in more detail.

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A. APPENDIX

A.1 LINEAR PROGRAMMING FORMULATION

The linear programming formulation of the platform's problem in Lemma 4 is given in the following. For $\alpha(\theta_i^k, \theta_j^{-k}) \in \{0, 1\}$, the platform's optimization problem can be represented by the following (mixed integer) linear program:

$$\max_{\{\Phi(\cdot), m(\cdot)\}_{ij}^k} \sum_{k=A,B} \sum_{\theta_i^k \in \Theta^k} \frac{(1-\delta)\nu(s_i^k)}{1-\rho} f(\theta_i^k), \quad (6)$$

subject to participation constraints $\forall \theta_i^k \in \Theta^k, k = A, B$,

$$\beta_i^k \omega_i^k \leq f(\theta_i^k)(\delta \omega_i^k - (1-\delta)s_i^k) + (1-\delta) \sum_j m(\theta_i^k, \theta_j^{-k}) \theta_i^k \theta_j^{-k}, \quad (7)$$

incentive constraints $\forall \theta_i^k \in \Theta^k, k = A, B$,

$$\begin{aligned} \beta_i^k \theta_i^k \theta_j^{-k} + \alpha(\theta_i^k, \theta_j^{-k})(-\beta_i^k \theta_i^k \theta_j^{-k}) &\leq f(\theta_i^k)(\delta \omega_i^k - (1-\delta)s_i^k) + (1-\delta) \sum_j m(\theta_i^k, \theta_j^{-k}) \theta_i^k \theta_j^{-k} \\ &\leq \left(\frac{\beta_i^k}{\delta} \theta_i^k \theta_j^{-k} - \beta_i^k \theta_i^k \theta_j^{-k} \right) (1 - \alpha(\theta_i^k, \theta_j^{-k})) + \beta_i^k \theta_i^k \theta_j^{-k}, \end{aligned} \quad (8)$$

feasibility and steady state constraints

$$\sum_{\theta_j^{-k} \in \Theta^{-k}} \Phi(\theta_i^k, \theta_j^{-k}) + \mathbf{1}_{k=B} \Phi(\theta_i^k, \omega_i^k) = f(\theta_i^k), \forall \theta_i^k \in \Theta^k, k = A, B, \quad (9)$$

$$f(\theta_i^k) = \frac{\beta_i^k - (1-\delta) \sum_j m(\theta_i^k, \theta_j^{-k})}{\delta}, \forall \theta_i^k \in \Theta^k, k = A, B, \quad (10)$$

and the following constraints on the matched and recommended pairs $\forall (\theta_i^k, \theta_j^{-k}) \in \Theta^k \times \Theta^{-k}$. First, the mass of recommended and matched pairs must be non-negative and the mass of matched pairs cannot be greater than the mass of recommended pairs

$$\Phi(\theta_i^k, \theta_j^{-k}) \geq 0, m(\theta_i^k, \theta_j^{-k}) \geq 0, \quad (11)$$

$$m(\theta_i^k, \theta_j^{-k}) \leq \Phi(\theta_i^k, \theta_j^{-k}). \quad (12)$$

Second, the mass of matched pairs must be smaller than the largest possible mass of the agents, i.e. the mass that arises when agents only exit upon becoming inactive β_i^k/δ times the acceptance probability, and larger than the mass of recommended pairs minus the

largest possible mass times the probability of a rejection

$$m(\theta_i^k, \theta_j^{-k}) \leq \frac{\min\{\beta_i^k, \beta_j^{-k}\}}{\delta} \alpha(\theta_i^k, \theta_j^{-k}), \quad (13)$$

$$m(\theta_i^k, \theta_j^{-k}) \geq \Phi(\theta_i^k, \theta_j^{-k}) - \frac{\min\{\beta_i^k, \beta_j^{-k}\}}{\delta} (1 - \alpha(\theta_i^k, \theta_j^{-k})). \quad (14)$$

This ensures that the mass of matched pairs does not exceed the mass of recommended pairs, and that if $\alpha(\theta_i^k, \theta_j^{-k}) = 0$ the mass of matched pairs is zero. To accommodate for mixed acceptance probabilities of agents, consider an agent of type θ_m^k that is indifferent between accepting and rejecting a type θ_s^{-k} . Hence, θ_m^k randomizes acceptance towards type θ_s^{-k} : $\sigma_k(\theta_m^k, \theta_s^{-k}) \in (0, 1)$. Conceptually, this imposes indifference or equality on some constraints rather than inequalities in the original formulation above. For any pair $(\theta_m^k, \theta_s^{-k}) \in \Theta^k \times \Theta^{-k}$ for which $\alpha(\theta_m^k, \theta_s^{-k}) \in (0, 1)$, the adjusted incentive constraints are

$$\beta_m^k \theta_m^k \theta_s^{-k} = f(\theta_m^k)(\delta \omega_m^k - (1 - \delta)s_m^k) + (1 - \delta) \sum_j m(\theta_m^k, \theta_j^{-k}) \theta_m^k \theta_j^{-k}, \text{ for } \theta_m^k, \quad (15)$$

$$\beta_s^{-k} \theta_m^k \theta_s^{-k} \geq f(\theta_s^{-k})(\delta \omega_s^{-k} - (1 - \delta)s_s^{-k}) + (1 - \delta) \sum_j m(\theta_m^k, \theta_j^{-k}) \theta_m^k \theta_j^{-k}, \text{ for } \theta_s^{-k}, \quad (16)$$

where θ_m^k is indifferent between accepting and rejecting θ_s^{-k} and θ_s^{-k} (weakly) accepts θ_m^k . The constraints on the mass of recommended and matched pairs are

$$m(\theta_m^k, \theta_s^{-k}) \leq \frac{\min\{\beta_m^k, \beta_s^{-k}\}}{\delta}, \text{ for } (\theta_m^k, \theta_s^{-k}), \quad (17)$$

$$m(\theta_m^k, \theta_j^{-k}) \leq \Phi(\theta_m^k, \theta_j^{-k}), \text{ for } (\theta_m^k, \theta_j^{-k}). \quad (18)$$

The linear program can be summarized in the subsequent lemma.

Lemma 6 (Linear Program). *Fix any mutual acceptance matrix A . The platform's maximization problem yields the same profit as linear programming problem with objective function in Equation 6 subject to constraints Equation 7 through 11 for any $\alpha(\theta_i^k, \theta_j^{-k}) \in \{0, 1\}$; and if $\alpha(\theta_m^k, \theta_s^{-k}) \in (0, 1)$ for any pair $(\theta_m^k, \theta_s^{-k}) \in \Theta^k \times \Theta^{-k}$, then for θ_m^k replace Equation 8 by Equation 15 and for θ_s^{-k} replace Equation 8 by Equation 16 as well as replace Equations 13 to 14 by Equations 17 to 18 for $(\theta_m^k, \theta_s^{-k})$.*

Note on Standard Form of a Linear Program To abbreviate future arguments, I express the linear program in standard form. The matrix notation is $\max x c^T$ subject to $Hx \leq b, x \geq 0$, where $c \in \mathbb{R}^n$. The variable vector $x \in X \subset \mathbb{R}^n$ consists of n variables, and X is compact. The m inequalities are given by matrix $H \in \mathbb{R}^{m \times n}$ with equalities expressed as two opposite inequalities. Vector $b \in \mathbb{R}^m$ captures the right-hand side of the inequalities. $\mathcal{P} \equiv \{x \in \mathbb{R}^n | Hx \leq b\}$ is the feasible region given by the inequality constraints.

B. APPENDIX: OMITTED PROOFS

Proof of Lemma 1 and 2 in the text.

Proof of Lemma 3 If $\eta_i^k < 1$ and $\Phi(\theta_i^k, \omega_i^k) \geq 0$ are optimal for any $\theta_i^k \in \Theta^k$, then $\eta_i^k = 1$ and $\Phi'(\theta_i^k, \omega_i^k)$ are also optimal such that

$$\Phi(\theta_i^k, \theta_j^{-k}) = \Phi'(\theta_i^k, \theta_j^{-k}), \quad (19)$$

$$(1 - \eta_i^k)f(\theta_i^k) + \Phi(\theta_i^k, \omega_i^k) = \Phi'(\theta_i^k, \omega_i^k), \quad (20)$$

for all $\theta_i^k \in \Theta^k$ and $\theta_j^{-k} \in \Theta^{-k}$. For given $\eta_i^k < 1$ and matching rule $\mathcal{M}$, Equation 19 and 20 determine the new matching rules for $\eta_i^k = 1$.

Now consider the participation for type θ_i^k . In equilibrium, the participation constraint must be binding for agents to find it optimal to randomize in their participation decision. Suppose the participation constraint is binding, then it can be rewritten as

$$(1 - \delta)s_i^k = (1 - \delta) \sum_j \alpha(\theta_i^k, \theta_j^{-k}) \phi(\theta_j^{-k} | \theta_i^k) (\theta_i^k \theta_j^{-k} - \omega_i^k).$$

As the masses are the same by Equation 19, the total surplus extracted by the platform remains the same as optimality requires that the participation constraint continues to bind. Multiplying with the total mass of agents of type θ_i^k if $\eta_i^k < 1$ yields

$$(1 - \delta)\eta_i^k f(\theta_i^k) s_i^k = (1 - \delta) \sum_j \alpha(\theta_i^k, \theta_j^{-k}) \Phi(\theta_j^{-k} | \theta_i^k) (\theta_i^k \theta_j^{-k} - \omega_i^k).$$

Similarly, when multiplying with the total mass of agents of type θ_i^k if $\eta_i^k = 1$ yields

$$(1 - \delta)f'(\theta_i^k) s_i^{k'} = (1 - \delta) \sum_j \alpha(\theta_i^k, \theta_j^{-k}) \Phi(\theta_j^{-k} | \theta_i^k) (\theta_i^k \theta_j^{-k} - \omega_i^k).$$

Therefore, the total surplus extracted is the same in both cases by construction. Thus, if the platform charges a search fee both cases yield the same surplus. In the case of advertising note that $f'(\theta_i^k)$ must increase if η_i^k increases, i.e. the steady-state mass increases if more agents participate everything else equal. Rewrite equation 19 as

$$\eta_i^k f(\theta_i^k) \phi(\theta_j^{-k} | \theta_i^k) = f'(\theta_i^k) \phi'(\theta_j^{-k} | \theta_i^k)$$

Therefore, to fulfill the equality in Equation 19 $\phi'(\theta_j^{-k} | \theta_i^k)$ must decrease to decrease the right-hand side. This implies that $s_i^{k'} < s_i^k$ and therefore, the platform profit increases in the advertising case due to the concavity of $\nu(s_i^k)$. $\square$

Proof of Lemma 5 As defined in the Section 4.2, the set $\mathcal{G}$ is the set of profit levels following from all linear programs with $A \in \mathcal{A}^*$. I show that the set $\mathcal{G}$ is (a) non-empty with $\Pi(A) < \infty$ for all $A \in \mathcal{A}^*$ and $-\infty < \Pi(A)$ for at least one $A \in \mathcal{A}^*$ and (b) finite.

To define set $\mathcal{G}$, recall the following definitions from the text. (i) Define a subset $\mathcal{A}^* \subset \mathcal{A}$, where $\mathcal{A}$ are the mutual acceptance matrices that can be implemented by a matching mechanism $\mathcal{M}$. Construct $\mathcal{A}^*$ through the following procedure: For every $A' \in \mathcal{A}$, construct a matrix A'' such that

$$\begin{aligned}\alpha'(\theta_i^k, \theta_j^{-k}) &= \alpha''(\theta_i^k, \theta_j^{-k}) \text{ if } \alpha'(\theta_i^k, \theta_j^{-k}) \in \{0, 1\}, \\ \alpha'(\theta_i^k, \theta_j^{-k}) &= \alpha_{ij} \text{ otherwise,}\end{aligned}$$

where α_{ij} is a variable in $[0, 1]$. (ii) For each $A \in \mathcal{A}^*$, the linear program is given by Lemma 6. The value of the objective is given by $\Pi(A)$. Then, (iii) $\mathcal{G} = \bigcup_{A \in \mathcal{A}^*} \Pi(A)$.

(a) $\mathcal{G}$ is non-empty. I will show that for any $A \in \mathcal{A}^*$, there exists an optimal value $\Pi(A) < \infty$ to the linear program. To do so, fix $A \in \mathcal{A}^*$ and consider the linear program as defined in Lemma 6 in Appendix A.1. To prove that an optimal solution exists, I show that: (i) the objective of the linear program is bounded, i.e., the linear program is not unbounded, and (ii) the feasible region of the variable vector, $\mathcal{P}$, is non-empty for a range of parameters. From both it follows that there exists an optimal solution by Dantzig (1963); Bertsimas and Tsitsiklis (1997).

(i) For fix $A \in \mathcal{A}^*$, the maximization problem is bounded if there exists a constant $C \in \mathbb{R}$ such that for all feasible $x \in \mathbb{R}^n$ $c^T x \leq C$ holds. The objective is bounded as

$$\sum_{k=A,B} \sum_{\theta_i^k \in \Theta^k} \frac{(1-\delta)s_i^k}{(1-\rho)} f(\theta_i^k) < \sum_{k=A,B} \sum_{\theta_i^k \in \Theta^k} \frac{(1-\delta)s_i^k}{(1-\rho)} \frac{\beta_i^k}{\delta} \equiv C. \quad (21)$$

This implies that $\Pi(A) < \infty$ for all $A \in \mathcal{A}^*$.

(ii) The feasible region is defined by the set $\mathcal{P} = \{x \in \mathbb{R}^n : Hx \leq b\}$. For any $A \in \mathcal{A}^*$, there exists a matching rule under which the constraints are not inconsistent for a range of parameters. This follows from the fact that $\mathcal{A}^* \subset \mathcal{A}$ and the definition of $\mathcal{A}$ implies that $A \in \mathcal{A}$ if and only if there exists an exogenous matching rule for which an equilibrium with mutual acceptance matrix A exists. By Lemma 2 there exists at least one equilibrium that can be implemented by a matching mechanism, hence, $\mathcal{A}^*$ is non-empty. Therefore, the feasible region is non-empty for a range of parameters for each linear program for fix $A \in \mathcal{A}^*$. Then, by strong duality (Dantzig, 1963), it follows that the linear program attains an optimal solution for any $A \in \mathcal{A}^*$. The optimal value to the linear program, $\Pi(A)$, is finite and $\mathcal{G}$ is non-empty.

(b) $\mathcal{G}$ is finite. As $\mathcal{G} = \bigcup_{A \in \mathcal{A}^*} \Pi(A)$ and $\mathcal{A}^*$ is finite by construction, $\mathcal{G}$ is also finite

as the profit level of a given linear program is a singleton. As each linear program for fix $A \in \mathcal{A}^*$ is bounded, the profit level takes on either a (finite) optimal value if an optimal solution exists or the value is undefined if the linear program is infeasible for given parameters. $\square$

Auxiliary Results for Theorem 1 To establish Theorem 1, I proceed through a sequence of intermediate lemmas, which are stated and proved below. Theorem 1 follows as a direct consequence. Recall that $s_i^k \in [0, \bar{u}] \equiv \mathcal{S}$ and denote $(s_1^k, \dots, s_N^k)^{k=A,B} \equiv \mathbf{s}$.

Lemma 7. *Let $\mathbf{s}$ be given. There exists an optimal solution with $\Pi^* \equiv \max \mathcal{G}(\mathbf{s})$.*

Proof. By Lemma 5, the set $\mathcal{G}$ is finite and non-empty for any given vector $\mathbf{s}$. Hence, $\mathcal{G}$ has a maximum element and $\Pi^* = \max \mathcal{G}$ is well-defined and has a finite value. $\square$

Let the platform choose the vector of search costs $\mathbf{s}$. Recall that $\mathcal{G}(\mathbf{s})$ denotes the set of profit levels induced by all feasible linear programs given $\mathbf{s}$.²¹ In slight abuse of notation, define $\mathcal{G}(\mathbf{s})$ as a correspondence from $\mathbf{s}$ to such profit levels $\Pi(\mathbf{s})$

$$\mathcal{G}(\mathbf{s}) : \mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|} \rightrightarrows \mathbb{R}_0^+.$$

which assigns to each point $\mathbf{s}$ of $\mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$ a finite subset $\mathcal{G}(\mathbf{s})$ of $\mathbb{R}_0^+$. The correspondence is compact-valued as $\mathcal{G}(\mathbf{s})$ is a compact (finite) subset of $\mathbb{R}_0^+$ for all $\mathbf{s} \in \mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$. In the following, I will show that the correspondence is upper hemicontinuous in $\mathbf{s}$ on $\mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$.

To do so, recall the matrix notation of the linear program in Appendix A.1:

$$\max_{x \in X} xc^T \equiv \Pi_A(\mathbf{s}), \text{ s.t. } H_A x \leq b_A, x \geq 0.$$

Denote by subscript A , the profit level and constraint set of the linear program for given matrix $A \in \mathcal{A}^*$. In Lemma 5, I have shown that a linear program for a fixed $A \in \mathcal{A}^*$ has a solution for some $\mathbf{s} \in \mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$. Additionally, whenever the linear program has a solution, it has an optimal solution. The value of the linear program, $\Pi_A(\mathbf{s})$, is thus finite on a set $\mathcal{J}_A \equiv \{\mathbf{s} \in \mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|} \mid -\infty < \Pi_A(\mathbf{s}) < \infty\}$, where $\mathcal{J}_A \subseteq \mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$. The set is compact.²²

²¹If s_i^k exceeds the maximal utility that type θ_i^k can achieve on the platform, they will not participate. Hence, if s_i^k exceeds the maximal utility $\forall \theta_i^k \in \Theta^k$ and $k = A, B$, the equilibrium profit is zero.

²²The set $\mathcal{J}_A$ contains all $\mathbf{s} \in \mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$ for which the value of the linear program is finite. In other words, the linear program must be bounded and feasible for those $\mathbf{s}$. By Lemma 5, the linear program is bounded. The linear program is feasible for some $\mathbf{s}$ if all constraints can be met, i.e. the feasible region $\mathcal{P}$ is non-empty. Suppose for contradiction that $\mathcal{J}_A$ is not compact. Now, take any sequence $\mathbf{s}_n \rightarrow \mathbf{s}$, for which the feasible region is non-empty for all $\mathbf{s}_n$. For the limit point $\mathbf{s}$ not to be in set $\mathcal{J}_A$, the feasible region must be empty for $\mathbf{s}$, and hence, at least one inequality must be violated strictly. But then, as the linear constraints are continuous in $\mathbf{s}$, the constraints must also be violated for $\mathbf{s}_n$ close enough to $\mathbf{s}$, a contradiction.

Lemma 8. *The value of the objective $\Pi_A(s)$ of a linear program for given matrix $A \in \mathcal{A}^*$ is upper hemicontinuous in $\mathbf{s}$ on $\mathcal{J}_A$.*

Proof. Fix $A \in \mathcal{A}^*$, and consider the associated linear program from Lemma 6. For given $A \in \mathcal{A}^*$, $\mathbf{s}$ changes vector c continuously, as each entry, $\nu(s_i^k)$ or 0, is continuous in s_i^k . Furthermore $\mathbf{s}$ changes matrix H_A continuously as s_i^k linearly enters as a coefficient in the incentive and participation constraints. The optimal value of the linear program is given by

$$\Pi_A(\mathbf{s}) \equiv \sup_{x \in \mathbb{R}^n} \{c(\mathbf{s})x \mid H_A(\mathbf{s})x \leq b_A, x \geq 0\},$$

which is finite on $\mathcal{J}_A$. In slight abuse of notation, denote the correspondence from $\mathbf{s}$ to the optimal value of the linear program by $\Pi_A(\mathbf{s}) : \mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|} \rightrightarrows \mathbb{R}_0^+$. Next, consider the set of primal feasible solutions of the linear program, $P_A(\mathbf{s})$, that defines objective Π . This is given by the correspondence $\mathbf{s} \rightarrow P_A(\mathbf{s}) \equiv \{x \mid H_A(\mathbf{s})x \leq b, x \geq 0\}$.

First, I show that the set of (primal) feasible solutions of the linear program is upper hemicontinuous in $\mathbf{s}$. Consider the following definition: $P_A(\mathbf{s})$ is upper hemicontinuous at $\mathbf{s}$ on $\mathcal{J}_A$ if

$$\mathbf{s} = \lim_{n \rightarrow \infty} \mathbf{s}_n, \quad x_n \in P_A(\mathbf{s}_n), \quad \text{and} \quad x = \lim_{n \rightarrow \infty} x_n,$$

implies that $x \in P_A(\mathbf{s})$.²³ To see that $P_A(\mathbf{s})$ is upper hemicontinuous, suppose that $\{\mathbf{s}_n\}_n \in \mathcal{J}_A$ and $\mathbf{s} = \lim_{n \rightarrow \infty} \mathbf{s}_n$. Let $\{x_n\}_n$ be a sequence such that for all n , $x_n \in P_A(\mathbf{s}_n)$: $H_A(\mathbf{s}_n)x_n \leq b_A$, and $x = \lim_{n \rightarrow \infty} x_n$. Since by the continuity of $H_A(\cdot)$ and independence of b_A in $\mathbf{s}$

$$\|H_A(\mathbf{s}_n) - H_A(\mathbf{s})\| \rightarrow 0, \quad \|x_n - x\| \rightarrow 0, \quad \text{and} \quad \|b_A - b_A\| = 0,$$

it follows that $H_A x \leq b_A$ and $x \geq 0$, which yields $x \in P_A(\mathbf{s})$. This implies that $P_A(\mathbf{s})$ is in fact upper hemicontinuous in $\mathbf{s}$ on $\mathcal{J}_A$. Next, I show that this implies that $\Pi_A(\mathbf{s}) = c(\mathbf{s})x$ is upper hemicontinuous in $\mathbf{s}$ on $\mathcal{J}_A$. Suppose that $\{\mathbf{s}_n\}_n \in \mathcal{J}_A$ and $\mathbf{s} = \lim_{n \rightarrow \infty} \mathbf{s}_n$. Let $\{\Pi_n\}_n$ be a sequence such that for all n , $\Pi_n \in \Pi_A(\mathbf{s}_n)$, and $\Pi = \lim_{n \rightarrow \infty} \Pi_n$. Since by the continuity of $c(\cdot)$, $\|c(\mathbf{s}_n) - c(\mathbf{s})\| \rightarrow 0$, and the upper hemicontinuity of $P_A(\mathbf{s})$ on $\mathcal{J}_A$, $\|x_n - x\| \rightarrow 0$, it follows that $\Pi \in \Pi_A(\mathbf{s})$. This implies that $\Pi_A(\mathbf{s})$ is in fact upper hemicontinuous in $\mathbf{s}$ on $\mathcal{J}_A$. $\square$

Lemma 9. $\mathcal{G}(\mathbf{s})$ is upper hemicontinuous in $\mathbf{s}$ on $\mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$.

Proof. Recall that $\mathcal{G}(\mathbf{s}) = \bigcup_{A \in \mathcal{A}^*} \Pi_A(\mathbf{s})$ is the finite union over the equilibrium profit levels of each linear program. For each $\Pi_A(\mathbf{s})$ the value $\Pi_A(\mathbf{s})$ is finite on $\mathcal{J}_A$ and

²³This definition follows Wets (1985). Furthermore, let $\|H\| = \sup_{x \in X} \|Hx\|$

empty on $\mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|} \setminus \mathcal{J}_A$. I prove the lemma by induction over the equilibria associated with the finite set $\mathcal{A}^*$. Let there be $\overline{K}$ equilibria, which can be implemented by the linear programs and consider the correspondence $\mathcal{G}_K(\mathbf{s}) = \bigcup_{\{A_1, \dots, A_K\}} \Pi_A(\mathbf{s})$ that includes K out of $\overline{K}$ equilibria. By induction, I will consider $\mathcal{G}_K$ to include increasingly more equilibria. **Base case:** Let $\mathcal{G}_1$ be the correspondence that includes only the trivial equilibrium from Lemma 2 with $A_1 \in \mathcal{A}^*$. Note that $\mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|} = \mathcal{J}_{A_1}$ as the trivial equilibrium is a solution to the corresponding linear program for each $\mathbf{s} \in \mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$. Hence, the statement follows from Lemma 8.

Induction step: The induction hypothesis states: $\mathcal{G}_K(\mathbf{s}) = \bigcup_{\{A_1, \dots, A_K\}} \Pi_A(\mathbf{s})$ is upper hemicontinuous on $\mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$. Note that by the induction step, K includes the trivial equilibrium. It remains to show that $\mathcal{G}_K(\mathbf{s}) \cup \Pi_{A_{K+1}}(\mathbf{s})$ is upper hemicontinuous in $\mathbf{s}$ on $\mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$. Recall that the correspondence $\mathcal{G}_K(\mathbf{s}) \cup \Pi_{A_{K+1}}(\mathbf{s})$ is upper hemicontinuous at $\mathbf{s}_0 \in \mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$, if for any open set $V \subseteq \mathbb{R}_0^+$ with $\mathcal{G}_K(\mathbf{s}_0) \cup \Pi_{A_{K+1}}(\mathbf{s}_0) \subseteq V$, there exists an open neighborhood $U(\mathbf{s}_0) \subseteq \mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$ such that if $\mathbf{s} \in U(\mathbf{s}_0)$, then $\mathcal{G}_K(\mathbf{s}) \cup \Pi_{A_{K+1}}(\mathbf{s}) \subseteq V$.

Let $\mathbf{s}_0 \in \mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$ and V be an open set with $\mathcal{G}_K(\mathbf{s}_0) \cup \Pi_{A_{K+1}}(\mathbf{s}_0) \subseteq V$. Suppose first that $\Pi_{A_{K+1}}$ is empty at $\mathbf{s}_0$. Since $\mathcal{G}_K(\mathbf{s}_0) \cup \Pi_{A_{K+1}}(\mathbf{s}_0) \subseteq V$, it follows that $\mathcal{G}_K(\mathbf{s}_0) \subseteq V$ and $\Pi_{A_{K+1}}(\mathbf{s}_0) \subseteq V$ by assumption (where V is the union of an open set and the empty set). By the upper hemicontinuity of $\mathcal{G}_K(\mathbf{s})$, there exists a neighborhood U_K of $\mathbf{s}_0$ such that $\mathcal{G}_K(\mathbf{s}_0) \subseteq V$ for all $\mathbf{s} \in U_K$. Additionally, there exists a neighborhood U_{K+1} of $\mathbf{s}_0$ such that $\Pi_{A_{K+1}}(\mathbf{s}_0) = \emptyset \subseteq V$ for all $\mathbf{s} \in U_{K+1}$ (by the compactness of $\mathcal{J}_{A_{K+1}}$. Let $U = U_K \cap U_{K+1}$. Then, for any $\mathbf{s} \in U$, both $\mathcal{G}_K(\mathbf{s}) \subseteq V$ and $\Pi_{A_{K+1}}(\mathbf{s}) \subseteq V$ such that $\mathcal{G}_K(\mathbf{s}) \cup \Pi_{A_{K+1}}(\mathbf{s}) \subseteq V$.

Let both $\mathcal{G}_K(\mathbf{s})$ and $\Pi_{A_{K+1}}(\mathbf{s})$ be non-empty at $\mathbf{s}_0$. Since $\mathcal{G}_K(\mathbf{s}_0) \cup \Pi_{A_{K+1}}(\mathbf{s}_0) \subseteq V$, it follows that $\mathcal{G}_K(\mathbf{s}_0) \subseteq V$ and $\Pi_{A_{K+1}}(\mathbf{s}_0) \subseteq V$. As both $\mathcal{G}_K(\mathbf{s}_0)$ and $\Pi_{A_{K+1}}(\mathbf{s}_0)$ are upper hemicontinuous for $\mathbf{s}_0$, it holds that: There exists a neighborhood U_K of $\mathbf{s}_0$ such that $\mathcal{G}_K(\mathbf{s}_0) \subseteq V$ for all $\mathbf{s} \in U_K$ and U_{K+1} of $\mathbf{s}_0$ such that $\Pi_{A_{K+1}}(\mathbf{s}_0) \subseteq V$ for all $\mathbf{s} \in U_{K+1}$. Then, for any $\mathbf{s} \in U$, both $\mathcal{G}_K(\mathbf{s}) \subseteq V$ and $\Pi_{A_{K+1}}(\mathbf{s}) \subseteq V$ such that $\mathcal{G}_K(\mathbf{s}) \cup \Pi_{A_{K+1}}(\mathbf{s}) \subseteq V$. Therefore, $\mathcal{G}(\mathbf{s})$ is upper hemicontinuous in $\mathbf{s}$ on $\mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$. $\square$

Lemma 10. *The function $\Pi^*(\mathbf{s})$ is upper semi-continuous in $\mathbf{s}$ on $\mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$.*

Proof. The function Π^* is upper-semicontinuous if for every point $\mathbf{s} \in \mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$, $\Pi(\mathbf{s}) \geq \limsup \Pi(\mathbf{s}_n)$ for every sequence $\{\mathbf{s}_n\}_n \subset \mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$ satisfying $\lim_{n \rightarrow \infty} \mathbf{s}_n = \mathbf{s}$.

Let $\lim_{n \rightarrow \infty} \mathbf{s}_n = \mathbf{s}$, and define $\Pi_n^* = \max \mathcal{G}(\mathbf{s}_n)$, so that $\Pi_n^* \in \mathcal{G}(\mathbf{s}_n)$ for all n . Since for each $\mathbf{s}_n$ $\mathcal{G}(\cdot)$ is finite by Lemma 5 and the sequence $\{\Pi_n^*\}$ is bounded, it has a convergent subsequence by the Bolzano-Weierstrass theorem: $\Pi_{n_k}^* \rightarrow \Pi'$ for some $\Pi' \in \mathbb{R}_0^+$. Then, as $\Pi_{n_k}^* \in \mathcal{G}(\mathbf{s}_{n_k})$, $\mathbf{s}_{n_k} \rightarrow \mathbf{s}$, and $\Pi_{n_k}^* \rightarrow \Pi'$, the upper hemicontinuity of $\mathcal{G}(\mathbf{s})$ implies that any limit point of $\Pi_{n_k}^*$ belongs to $\mathcal{G}(\mathbf{s})$, i.e. $\Pi' \in \mathcal{G}(\mathbf{s})$. Therefore, $\Pi' \leq \max \mathcal{G}(\mathbf{s})$. Since

$\Pi_{n_k}^* \rightarrow \Pi'$, this implies:

$$\lim_{n \rightarrow \infty} \sup \Pi_n = \lim_{n \rightarrow \infty} \sup \max \mathcal{G}(\mathbf{s}_n) \leq \max \mathcal{G}(\mathbf{s}). \quad \square$$

Proof of Theorem 1 By Lemma 10, $\max \mathcal{G} = \Pi^*(\mathbf{s})$ is upper semi-continuous in $\mathbf{s}$ and compact-valued. Thus, there exists a maximum by Weierstrass extreme value theorem on the compact set $\mathcal{S}^{|\Theta^k| \times |\Theta^{-k}|}$. $\square$

Proof of Proposition 1 The proof proceeds by considering the cases where search costs are exogenous and where search costs are chosen as search fee or advertising.

Case 1: Exogenous Search Cost First, suppose search costs are exogenously given. Let the parameters be drawn uniformly from the following sets: $\theta_i^k \in \Theta^k = [\underline{\theta}, \bar{\theta}] \subseteq \mathbb{R}_+$, $\beta_i^k \in [0, \bar{\beta}]$, $\delta \in (0, 1]$, $\omega_i^k \in \Omega = [0, \bar{\omega}]$, and $s_i^k \in [0, \bar{u}]$. An outcome is said to be generically suboptimal if the set of parameter values for which it is optimal has measure zero in the relevant parameter space.

For given $A \in \mathcal{A}^*$, an optimal solution is a matching rule for which the objective function of the linear program in Appendix A.1 attains its maximum value. The platform solves the linear program in Lemma 6. Both feasibility (Equation 9) and steady-state constraints (Equation 10) must be binding in the optimal solution. Additionally, at least one participation (Equation 7) or incentive constraint (Equation 8) must be binding in the optimal solution. Suppose otherwise, then the platform can decrease at least one $m(\theta_i^k, \theta_j^{-k})$, hence increase $f(\theta_i^k)$, such that one constraint is binding and thereby increase its profits.

Following Lauermann and Nöldeke (2014), $\{m^{RM}(\theta_i^k, \theta_j^{-k})\}_{ij}^k$ is the vector of masses of matched pairs under random matching. Then, $m^{RM}(\theta_i^k, \theta_j^{-k}) = 0$ if $\alpha(\theta_i^k, \theta_j^{-k}) = 0$ and

$$m^{RM}(\theta_i^k, \theta_j^{-k}) = \frac{\alpha(\theta_i^k, \theta_j^{-k}) \beta_i^k \mu(\theta_i^k, \omega_i^k) \beta_j^{-k} \mu(\theta_j^{-k}, \omega_j^{-k})}{\left(\sum_{\theta_i^k} \beta_i^k \mu(\theta_i^k, \omega_i^k) \right) \cdot \left(\sum_{\theta_j^{-k}} \beta_j^{-k} \mu(\theta_j^{-k}, \omega_j^{-k}) \right)}$$

if $\alpha(\theta_i^k, \theta_j^{-k}) \in (0, 1]$ (see Appendix C.1 for the extended analysis). This is a function of the inflow vector $(\beta_1^k, \dots, \beta_{N^k}^k)_k$, δ and the probability of θ_i^k being matched to their outside option ω_i^k ($\mu(\theta_i^k, \omega_i^k)$). Observe that for given $A \in \mathcal{A}^*$, $m^{RM}(\theta_i^k, \theta_j^{-k})$ is independent of s_i^k . Fix $A \in \mathcal{A}^*$. Given $\{m^{RM}(\theta_i^k, \theta_j^{-k})\}_{ij}^k$, the participation and incentive constraints are

generically non-binding. Rearranging and using the steady state condition yields

$$\frac{\beta_i^k}{1-\delta} \left(\theta_i^k \theta_j^k - \omega_i^k + \frac{(1-\delta)}{\delta} s_i^k \right) \leq \sum_j m^{RM}(\theta_i^k, \theta_j^{-k}) \left(\theta_i^k \theta_j^{-k} - \omega_i^k + \frac{(1-\delta)}{\delta} s_i^k \right), \quad (22)$$

$$\beta_i^k \frac{s_i^k}{\delta} \leq \sum_j m^{RM}(\theta_i^k, \theta_j^{-k}) \left(\theta_i^k \theta_j^{-k} - \omega_i^k + \frac{(1-\delta)}{\delta} s_i^k \right). \quad (23)$$

Suppose $\beta_i^k, \omega_i^k, \theta_i^k, \theta_j^{-k}$ and δ are drawn uniformly from their continuous intervals. Note that each constraint for a type θ_i^k is a linear equation in s_i^k . Hence, for given $\{m^{RM}(\theta_i^k, \theta_j^{-k})\}_{ij}^k$, there exists at most one s_i^k per participation or incentive constraint of type θ_i^k such that the constraint is binding. This implies that if s_i^k is drawn uniformly from a continuous interval, the set of parameters for which the constraint is binding has measure zero. Therefore integrating over the cases for which at least one constraint is binding, the corresponding set of parameters has measure zero as well. Hence, for each $A \in \mathcal{A}^*$, the constraints are generically non-binding. Lastly, since $\mathcal{A}^*$ is finite, this concludes the proof for exogenously given search costs.

Case 2: Endogenous Search Cost (Search Fee) With endogenous search fees and $\nu(s_i^k) = s_i^k$, the optimization problem is equivalent to a linear assignment problem that maximizes total match output over all possible pairings subject to feasibility constraints that ensure each agent can match at most once. Since the matching technology is super-modular, the solution is positive assortative matching (instead of random matching). For more details, see Appendix C.1. Given the solution to this problem, $\{m^{PAM}(\theta_i^k, \theta_j^{-k})\}_{ij}^k$, the platform sets s_i^k to fully extract each type's surplus:

$$\beta_i^k \frac{(1-\delta)}{\delta} s_i^k = (1-\delta) \sum_j m^{PAM}(\theta_i^k, \theta_j^{-k}) \left(\theta_i^k \theta_j^{-k} - \omega_i^k + \frac{(1-\delta)}{\delta} s_i^k \right).$$

Case 3: Endogenous Search Cost (Advertising) Now consider the case in which the platform earns a revenue of $\nu(s_i^k)$ when charging search costs s_i^k . Fix $A \in \mathcal{A}^*$ and let the random matching vector be given $\{m^{RM}(\theta_i^k, \theta_j^{-k})\}_{ij}^k$. Using the steady-state conditions to substitute for $f^{RM}(\theta_i^k)$, the platform's objective under random matching becomes the following maximization problem:

$$\max_{\mathbf{s}} \sum_{k=A,B} \sum_{\theta_i^k \in \Theta^k} \frac{(1-\delta)\nu(s_i^k)}{1-\rho} \underbrace{\frac{\beta_i^k - (1-\delta) \sum_j m^{RM}(\theta_i^k, \theta_j^{-k})}{\delta}}_{=f^{RM}(\theta_i^k)}. \quad (24)$$

subject to the participation and incentive constraints in Equation 22 and 23.

To maximize profit with respect to $\mathbf{s}$, observe first that $\nu'(s_i^k) > 0$ as $\nu(s_i^k)$ is strictly increasing in s_i^k . This implies that the platform has an incentive to increase the search

costs as much as possible given the constraints. Therefore for $A \in \mathcal{A}^*$, the optimal solution is to choose s_i^k such that for each type $\theta_i^k \in \Theta^k, k = A, B$ either the participation or the relevant incentive constraint induced by A is binding. Note that the random matching vector satisfies the feasibility condition, and as random matching is independent of s_i^k feasibility remains to be satisfied.

Next, I show that the platform has an incentive to deviate from the above solution. First, suppose $A' \in \mathcal{A}^*$ consists only of entries equal to one. Incentive constraints are slack, and the platform chooses $\mathbf{s}$ to make participation constraints binding. Under random matching, the platform can at most charge the expected value of a match. By deviating to PAM the platform can raise search costs and profits, since $\nu(s_i^k)$ is strictly increasing in s_i^k .

Second, consider any matrix in $A'' \in \mathcal{A}^* \setminus \{A'\}$, i.e., at least one type rejects another type with positive probability. Due to supermodularity, this implies that at least one type is willing to reject the lowest type. Consider the pair of types $(\theta_1^k, \theta_R^{-k})$ for which type $\theta_R^{-k} \in \Theta^{-k}$ is willing to reject the lowest type θ_1^k on the other market side ($\alpha(\theta_1^k, \theta_R^{-k}) = 0$). Recall that each type must be accepted by at least one other type on the opposite market side to be willing to participate, thus consider pairs $(\theta_1^k, \theta_A^{-k})$ and $(\theta_A^k, \theta_R^{-k})$ for which $\alpha(\theta_1^k, \theta_A^{-k}) = 1$ and $\alpha(\theta_A^k, \theta_R^{-k}) = 1$. For fix $A'' \in \mathcal{A}^* \setminus \{A'\}$, I will show that the platform's profit can be improved by changing the matching rules for types $\theta_1^k, \theta_A^k, \theta_A^{-k}$ and θ_R^{-k} as well as adjusting their search costs. The platform will choose the mass of recommended pairs $\Phi'(\theta_1^k, \theta_R^{-k}), \Phi'(\theta_1^k, \theta_A^{-k}), \Phi'(\theta_A^k, \theta_R^{-k}), \Phi'(\theta_A^k, \theta_A^{-k})$, and the mass of matched pairs $m'(\cdot, \cdot) = \alpha(\cdot, \cdot)\Phi'(\cdot, \cdot)$ as detailed below. For all other types, the platform chooses the mass of recommended pairs such that they equal the mass of recommended pairs under random matching: $\Phi'(\theta_i^k, \theta_j^{-k}) = \Phi^{RM}(\theta_i^k, \theta_j^{-k})$. Without loss of generality, suppose that the total mass of all types on market side A is smaller or equal than side B . Then, for market side A , the platform chooses the mass of types that are recommended to their outside option such that $\Phi'(\theta_i^A, \omega_i^A) = \Phi^{RM}(\theta_i^A, \omega_i^A)$ for all $\theta_i^A \in \Theta^A \setminus \{\theta_1^A, \theta_A^A\}$. The mass of recommended pairs and the mutual acceptance probabilities remain the same as under random matching. Therefore, the participation, incentive constraints, feasibility constraints (Equation 9) and steady-state constraints (Equation 10) for all other types continue to hold.

For $\varepsilon \in [-\min\{\beta_i - m^{RM}(\cdot, \cdot)\}, \min\{m^{RM}(\cdot, \cdot)\}]$, the platform chooses $m^{RM}(\theta_1^k, \theta_A^{-k}) - m'(\theta_1^k, \theta_A^{-k}) = \varepsilon$ and $m^{RM}(\theta_A^k, \theta_R^{-k}) - m'(\theta_A^k, \theta_R^{-k}) = \varepsilon$, i.e. the platform changes the mass of the two matched pairs by ε . Substituting the change into the steady state condition (Equation 10) for type θ_1^k and type θ_R^{-k} increases the steady state mass by $\frac{1-\delta}{\delta}\varepsilon$ for $\varepsilon > 0$, and decreases otherwise. Substituting $\Phi^{RM}(\theta_1^k, \theta_A^{-k}) - \Phi'(\theta_1^k, \theta_A^{-k}) = \varepsilon$ and $\Phi^{RM}(\theta_A^k, \theta_R^{-k}) - \Phi'(\theta_A^k, \theta_R^{-k}) = \varepsilon$ into the feasibility constraints (Equation 9) of type θ_1^k

and type θ_R^{-k} implies that $\Phi'(\theta_1^k, \theta_R^{-k}) - \Phi^{RM}(\theta_1^k, \theta_R^{-k}) = \frac{1-\delta}{\delta}\varepsilon + \varepsilon = \varepsilon/\delta$. It remains to determine $\Phi'(\theta_A^k, \theta_A^{-k}) - \Phi^{RM}(\theta_A^k, \theta_A^{-k})$ and $m'(\theta_A^k, \theta_A^{-k}) - m^{RM}(\theta_A^k, \theta_A^{-k})$. To do so, consider two cases: either $\alpha(\theta_A^k, \theta_A^{-k}) = 0$ or $\alpha(\theta_A^k, \theta_A^{-k}) = 1$.

In the first case, $\alpha(\theta_A^k, \theta_A^{-k}) = 0$, I can exchange θ_A^k for θ_1^k and θ_A^{-k} for θ_R^{-k} . Then, it follows that $\Phi'(\theta_A^k, \theta_A^{-k}) - \Phi^{RM}(\theta_A^k, \theta_A^{-k}) = \varepsilon/\delta$ and $m'(\theta_A^k, \theta_A^{-k}) = 0$. In the second case, $\alpha(\theta_A^k, \theta_A^{-k}) = 1$, the platform can set $\Phi'(\theta_A^k, \theta_A^{-k}) - \Phi^{RM}(\theta_A^k, \theta_A^{-k}) = m'(\theta_A^k, \theta_A^{-k}) - m^{RM}(\theta_A^k, \theta_A^{-k}) = \varepsilon$. Since the platform decreases (increases) the mass of the matched pair $(\theta_A^k, \theta_R^{-k})$ but increases (decreases) the mass of the matched pair $(\theta_A^k, \theta_A^{-k})$ by the same amount, this implies that the steady state mass of type θ_A^k is unchanged compared to the steady state masses under random matching. Additionally, feasibility continues to be satisfied as the platform shift mass ε from one recommended pair to the other. Similarly, the steady state mass of type θ_A^{-k} is the same as under random matching and the steady state constraint as well as feasibility constraint remain satisfied.

Next determine the change in search costs for types $\theta_1^k, \theta_A^k, \theta_A^{-k}$ and θ_R^{-k} . Note that for the newly chosen mass of recommended and matched pairs $(\Phi'(\cdot, \cdot), m'(\cdot, \cdot))$, the originally binding participation or incentive constraint is no longer binding. Since, however, the right-hand side of the participation or incentive constraints (see Equation 22 and 23) are linearly increasing in $m(\cdot, \cdot)$ and the left-hand side of the constraints are ordered due to the supermodularity of the match utility, the platform can choose a new search cost $\tilde{s}_i^k$ such that the constraint becomes binding again. Let the platform choose $\tilde{s}_1^k, \tilde{s}_A^k, \tilde{s}_R^{-k}, \tilde{s}_A^{-k}$ such that originally binding participation or incentive constraint of each type is binding again. Using Equations 22 and 23 and $m^{RM}(\theta_1^k, \theta_A^{-k}) - m'(\theta_1^k, \theta_A^{-k}) = \varepsilon$ and $m^{RM}(\theta_A^k, \theta_R^{-k}) - m'(\theta_A^k, \theta_R^{-k}) = \varepsilon$, the difference between $s_1^k - \tilde{s}_1^k$ and $s_R^{-k} - \tilde{s}_R^{-k}$ can be directly obtained. From there, it follows that $\tilde{s}_1^k$ must be smaller than s_1^k for $\varepsilon > 0$ and $s_R^{-k} > \tilde{s}_R^{-k}$ for $\varepsilon > 0$. Next, if $\alpha(\theta_A^k, \theta_A^{-k}) = 0$, the difference between the search costs for types θ_A^k and θ_A^{-k} can be derived as above and again, it holds that $s_A^k > \tilde{s}_A^k$ and $s_A^{-k} > \tilde{s}_A^{-k}$ for $\varepsilon > 0$. If $\alpha(\theta_A^k, \theta_A^{-k}) = 1$, recall that the platform sets: $m^{RM}(\theta_1^k, \theta_A^{-k}) - m'(\theta_1^k, \theta_A^{-k}) = \varepsilon$, $m^{RM}(\theta_A^k, \theta_R^{-k}) - m'(\theta_A^k, \theta_R^{-k}) = \varepsilon$ and $m^{RM}(\theta_A^k, \theta_A^{-k}) - m'(\theta_A^k, \theta_A^{-k}) = -\varepsilon$. Using Equations 22 and 23, the difference of $s_A^{-k} - \tilde{s}_A^{-k}$ can be obtained and $s_A^{-k} < \tilde{s}_A^{-k}$ for $\varepsilon > 0$. Similarly, the difference of $s_A^k - \tilde{s}_A^k$ can be calculated.

To determine whether the deviation is profitable, consider the difference in profits between the deviation profits and random matching profits (Equation 24). In the first case, when $\alpha(\theta_A^k, \theta_A^{-k}) = 0$, the steady state mass of all four types $(\theta_1^k, \theta_A^k, \theta_R^{-k}, \theta_A^{-k})$ increases (decreases) by $(1-\delta)\varepsilon/\delta$ while their search costs decrease (increase). Taking the difference in profits and differentiating with respect to ε_i^k , and evaluating the condition at $\varepsilon = 0$,

yields:

$$\sum_{\theta_i^k \in \{\theta_1^k, \theta_A^k, \theta_R^k, \theta_A^{-k}\}} \nu'(\tilde{s}_i^k)|_{\tilde{s}_i^k=s_i^k} \frac{\partial \tilde{s}_i^k}{\partial \varepsilon}|_{\varepsilon=0} f^{RM}(\theta_i^k) + \frac{1-\delta}{\delta} \nu(\tilde{s}_i^k)|_{\tilde{s}_i^k=s_i^k}. \quad (25)$$

For analytical convenience, consider the class of concave functions $\nu(s_i^k) = \kappa(s_i^k)^\alpha$ for $\kappa \in \mathbb{R}^+$ and $\alpha \in (0, 1)$ from now on. Substituting $\nu(s_i^k) = \kappa(s_i^k)^\alpha$, $\nu'(s_i^k) = \kappa\alpha(s_i^k)^{\alpha-1}$, and the total derivatives of the differences in search costs into Equation 25 yields expression D (see Appendix C.2).²⁴ For the deviation to be profitable, the expression must be non-zero when being evaluated at $\varepsilon = 0$. Then, since the function D is continuous in α , $D > 0$ for $\alpha = 0$, and D is increasing in α for $\alpha < \alpha''$ and decreasing for $\alpha > \alpha''$, it follows that D has at most one root. In the second case, when $\alpha(\theta_A^k, \theta_A^{-k}) = 1$, the steady state mass of types (θ_1^k, θ_R^k) increases (decreases) by $\frac{(1-\delta)\varepsilon}{\delta}$ while the steady state mass of types θ_A^k and θ_A^{-k} remains unchanged. Repeating the same steps as above yields expression D_2 , which again has at most one root.

Now, suppose $\beta_i^k, \omega_i^k, \theta_i^k, \theta_j^{-k}$ and δ are drawn uniformly from their continuous intervals. Then, there exists at most one α for which $D = 0$ (or $D_2 = 0$). Let α be drawn uniformly from $(0, 1)$, then random matching is generically suboptimal as such α is drawn with measure zero. $\square$

Proof of Proposition 2 Since market sides are fully symmetric, for brevity I drop the superscript k . PAM is defined as $\phi(\theta_i|\theta_j) = 1$ iff $i = j$ and results in $f(\theta_i) = \beta_i \forall \theta_i \in \Theta$.

Case 1: Search Fee with a Price Ceiling. (a) “If” direction: PAM is optimal if the platform sets $s_i = \theta_i^2 - \omega_i$ for all $\theta_i \in \Theta$. As shown in Appendix C.1, PAM maximizes total match surplus across all agents. By choosing $s_i = \theta_i^2 - \omega_i$, the platform can extract each agent’s match surplus, thereby maximizing the platform’s profit.

(b) “Only if” direction: Suppose, for contradiction, that PAM is profit-maximizing even if $s_i < \bar{s}_i$ for some $\theta_i \in \Theta \setminus \{\theta_1\}$ and $\bar{s}_i = \min\{\theta_i^2 - \omega_i, \theta_i^2 - \frac{\theta_i\theta_1 - \delta\omega_i}{1-\delta}\}$. Observe that if the platform uses PAM with probability one in the next period and $s_i < \bar{s}_i$, then type θ_i would reject the lowest type θ_1 in the (zero-probability) event they meet, since

$$\max\{\theta_i\theta_1, \omega_i\} < \delta\omega_i + (1-\delta)(\theta_i^2 - s_i). \quad (26)$$

Consider a deviation from PAM in which all types other than θ_1 and θ_i continue to only meet each other ($\Phi^D(\theta_j, \theta_j) = \beta_j$), but θ_1 and θ_i meet each other with mass $\epsilon \in (0, \min\{\beta_1, \beta_i\}/\delta]$ ($\Phi^D(\theta_i, \theta_1) = \epsilon$). Simultaneously, reduce the search fee of type θ_1 from $s_1 = \theta_1^2 - \omega_1$ to some s'_1 , which I will specify below. Then, I will show that there exists an

²⁴The addition to the proof of Proposition 1 with detailed mathematical steps for the remaining paragraph can be found in Appendix C.2.

$\epsilon > 0$ and a corresponding s'_1 such that the resulting matching rule is feasible, incentive compatible, and strictly improves the platform's profit. To check feasibility, substitute the steady state conditions in Equation 10 into the feasibility constraints in Equation 9 and solve for the new (conditional) matching probabilities under the deviation, ϕ^D :

$$\phi^D(\theta_i|\theta_i) = \frac{\beta_i - \epsilon\delta}{\beta_i + (1-\delta)\epsilon}, \quad \phi^D(\theta_1|\theta_1) = \frac{\beta_1 - \epsilon\delta}{\beta_1 + (1-\delta)\epsilon}. \quad (27)$$

For any $\min\{\beta_1, \beta_i\}/\delta > \epsilon > 0$, $1 > \phi^D(\cdot) > 0$. Set $s'_1 = \frac{\beta_1 - \epsilon\delta}{\beta_1 + (1-\delta)\epsilon}(\theta_1^2 - \omega_1)$. To verify that type θ_i continues to reject type θ_1 , note that under PAM, if $s_i < \bar{s}_i$, the inequality in Equation 8 is slack. Since matching probabilities are continuous in ϵ , there exists a small $\epsilon > 0$ such that the incentive condition remains non-binding or becomes just binding. Thus, search behavior does not change for sufficiently small ϵ . Now consider the PC of type θ_1 (Equation 7). Under PAM, it is binding if $s_1 = \theta_1^2 - \omega_1$. Since θ_1 now meets type θ_i with positive probability, continuing to charge $s_1 = \theta_1^2 - \omega_1$ would violate the constraint. By lowering the search fee to s'_1 as defined above, the constraint remains binding. The platform's profit given the new matching rule and search fee is

$$\Pi^D = \frac{2(1-\delta)}{1-\rho} \left((\beta_i + \epsilon)s_i + \frac{(\beta_1 + \epsilon)(\beta_1 - \epsilon\delta)}{\beta_1 + (1-\delta)\epsilon}(\theta_1^2 - \omega_1) + \sum_{j \neq 1, i} \beta_j(\theta_j^2 - \omega_j) \right).$$

The deviation is profitable if $\Pi^D - \Pi^{PAM} > 0$, that is if $s_i - \frac{\epsilon\delta}{\beta_1 + (1-\delta)\epsilon}(\theta_1^2 - \omega_1) > 0$ for ϵ small enough.

Case 2: Advertisement. Given PAM, the platform sets $s_i = \theta_i - \omega_i$:

$$\Pi^{PAM} = \frac{2(1-\delta)}{1-\rho} \sum_{\theta_i \in \Theta} \nu(\theta_i^2 - \omega_i)\beta_i.$$

Consider a deviation as in Case 1, in which all types other than type θ_1 and some type θ_i continue to meet only each other. For type θ_1 and θ_i choose the mass of recommended and matched pairs (denoted by a superscript prime) such that $\beta_1 - \Phi'(\theta_1, \theta_1) = \beta_1 m'(\theta_1, \theta_1) = \epsilon$ and $\beta_i - \Phi'(\theta_i, \theta_i) = \beta_i m'(\theta_i, \theta_i) = \epsilon$ for $\epsilon \in (0, \min\{\beta_1, \beta_i\}]$. The new matching rule must satisfy the feasibility constraints (Equation 9) and steady state conditions (Equation 10). It follows that $\Phi'(\theta_1, \theta_i) = \frac{\epsilon}{\delta}$. To ensure that type θ_i rejects type θ_1 under the new matching rule (so that $m'(\theta_1, \theta_i) = 0$), while type θ_1 participates, the platform chooses $(\tilde{s}_1, \tilde{s}_i)$ such that

$$\begin{aligned} \tilde{s}_i &\in \left\{ \tilde{s}_i \in \mathbb{R}_+ \left| \beta_i \left(\max\{0, \theta_i \theta_1 - \omega_i\} + \frac{(1-\delta)}{\delta} \tilde{s}_i \right) = (1-\delta)(\beta_i - \epsilon) \left(\theta_i^2 - \omega_i + \frac{(1-\delta)}{\delta} \tilde{s}_i \right) \right. \right\}, \\ \tilde{s}_1 &\in \left\{ \tilde{s}_1 \in \mathbb{R}_+ \left| \beta_1 \left(\frac{(1-\delta)}{\delta} \tilde{s}_1 \right) = (1-\delta)(\beta_1 - \epsilon) \left(\theta_1^2 - \omega_1 + \frac{(1-\delta)}{\delta} \tilde{s}_1 \right) \right. \right\}. \end{aligned}$$

The deviation profit is therefore:

$$\Pi^D = \frac{2(1-\delta)}{1-\rho} \left(\nu(\tilde{s}_1) \left(\beta_1 + \frac{1-\delta}{\delta} \varepsilon \right) + \nu(\tilde{s}_i) \left(\beta_i + \frac{1-\delta}{\delta} \varepsilon \right) + \sum_{\theta_j \in \Theta \setminus \{\theta_1, \theta_i\}} \nu(\theta_j^2 - \omega_j) \beta_j \right).$$

The deviation is profitable if $\Pi^D - \Pi^{PAM} > 0$, or equivalently

$$\delta \leq \frac{(\nu(\tilde{s}_1) + \nu(\tilde{s}_i))\varepsilon}{\beta_1(\nu(s_1) - \nu(\tilde{s}_1)) + \beta_i(\nu(s_i) - \nu(\tilde{s}_i)) + (\nu(\tilde{s}_1) + \nu(\tilde{s}_i))\varepsilon} \equiv \bar{\delta}, \text{ where } \bar{\delta} \in (0, 1) \text{ for } \varepsilon > 0. \quad \square$$

Case 3: Complexity Constrained Pricing. Select two types with a common search fee s , denoted $\theta_h > \theta_l$. Under PAM, $s \leq \theta_l^2 - \omega_l$ for high and low types to participate. Consider a deviation from PAM where all other types continue to only meet each other, $(\Phi^D(\theta_j, \theta_j) = \beta_j)$. For each possible acceptance matrix (see Section 4.4), solve the auxiliary problem in Appendix A.1. First, allow type-dependent search fees $s_h \neq s_l$.

(a) A_{PAM} : A_{PAM} induces the following incentive and participation constraints:

$$\beta_h \theta_h \theta_l \leq f(\theta_h)(\delta \omega_h - (1-\delta)s_h) + (1-\delta)\Phi(\theta_h, \theta_h)\theta_h^2, \quad (28)$$

$$\beta_l \omega_l \leq f(\theta_l)(\delta \omega_l - (1-\delta)s_l) + (1-\delta)\Phi(\theta_l, \theta_l)\theta_l^2. \quad (29)$$

As the platform maximizes the steady state mass, it minimizes $\Phi(\theta_h, \theta_h)$ and $\Phi(\theta_l, \theta_l)$ subject to these constraints. When both Equation 28 and Equation 29 bind,

$$\Phi^{(a)}(\theta_h, \theta_h) = \frac{\beta_h \left(\delta(\theta_h \theta_l - (s_h + \omega_h)) + s_h \right)}{(1-\delta) \left(\delta(\theta_h^2 - (s_h + \omega_h)) + s_h \right)}, \quad \Phi^{(a)}(\theta_l, \theta_l) = \frac{\beta_l s_l}{\delta(\theta_l^2 - (s_l + \omega_l)) + s_l}.$$

Both constraints can bind simultaneously only if

$$\left(\frac{\beta_h}{\beta_l} \right)^{(a)} = \frac{(1-\delta)(\delta(\theta_h^2 - s_h - \omega_h) + s_h)(\theta_l^2 - s_l - \omega_l)}{(\delta(\theta_l^2 - s_l - \omega_l) + s_l) \left(\theta_h(\theta_h - \theta_l) - s_h - \delta(\theta_h^2 - s_h - \omega_h) \right)},$$

due to the feasibility constraints, Equation 9. Otherwise, only one constraint binds. (i) If $\frac{\beta_h}{\beta_l} \geq \left(\frac{\beta_h}{\beta_l} \right)^{(a)}$, the participation constraint binds implying $\Phi(\theta_l, \theta_l) = \Phi^{(a)}(\theta_l, \theta_l)$. (ii) If $\frac{\beta_h}{\beta_l} \leq \left(\frac{\beta_h}{\beta_l} \right)^{(a)}$, the incentive constraint binds, implying $\Phi(\theta_h, \theta_h) = \Phi^{(a)}(\theta_h, \theta_h)$. By using the feasibility and steady-state constraints, the other masses can be calculated, with

$$f(\theta_i) = \frac{1}{\delta}(\beta_i - (1-\delta)\Phi(\theta_i, \theta_i)).$$

In (i), the profit is

$$\Pi_{A_{PAM}}^{(i)} = \frac{(1-\delta)\nu(s_h)}{1-\rho} f(\theta_h) + \frac{(1-\delta)\nu(s_l)}{1-\rho} f(\theta_l), \quad (30)$$

where

$$f(\theta_h) = \frac{(\theta_l^2 - s_l - \omega_l)(\beta_h - \beta_l)\delta + \beta_l(\theta_l^2 - s_l - \omega_l) + \beta_h s_l}{(\theta_l^2 - s_l - \omega_l)\delta + s_l}, f(\theta_l) = \frac{\beta_l(\theta_l^2 - \omega_l)}{(\theta_l^2 - s_l - \omega_l)\delta + s_l}.$$

In (ii), the profit is

$$\Pi_{APAM}^{(ii)} = \frac{(1 - \delta)\nu(s_h)}{1 - \rho} f(\theta_h) + \frac{(1 - \delta)\nu(s_l)}{1 - \rho} f(\theta_l), \quad (31)$$

where

$$f(\theta_h) = \frac{\beta_h \theta_h (\theta_h - \theta_l)}{(\theta_h^2 - s_h - \omega_h)\delta + s_h}, f(\theta_l) = \frac{\beta_h (\theta_h (\theta_h - \theta_l) - s_h) + \beta_l s_h - (\theta_h^2 - s_h - \omega_h)(\beta_h - \beta_l)\delta}{(\theta_h^2 - s_h - \omega_h)\delta + s_h}.$$

(b) A_{WPAM} : (b.1) A_{WPAM} induces the following constraints: A high type must be indifferent between searching and accepting low types, while the low type must be willing to participate. To maximize profits, the low type's participation constraint must bind as well. Otherwise, the platform must increase the mass of matches between the high and the low type, which decreases rent from both types for given search cost.

$$\beta_h \theta_h \theta_l = f(\theta_h)(\delta \omega_h - (1 - \delta)s_h) + (1 - \delta)(\Phi(\theta_h, \theta_h)\theta_h^2 + m(\theta_h, \theta_l)\theta_h \theta_l), \quad (32)$$

$$\beta_l \omega_l = f(\theta_l)(\delta \omega_l - (1 - \delta)s_l) + (1 - \delta)(\Phi(\theta_l, \theta_l)\theta_l^2 + m(\theta_h, \theta_l)\theta_h \theta_l). \quad (33)$$

Together with the feasibility and steady state constraints, the set of equations determines $\Phi^{(b)}(\theta_h, \theta_h)$, $\Phi^{(b)}(\theta_h, \theta_l)$, $\Phi^{(b)}(\theta_l, \theta_l)$, $m^{(b)}(\theta_h, \theta_l)$, where $\Phi^{(b)}(\theta_h, \theta_h) = m^{(b)}(\theta_h, \theta_h)$ and $\Phi^{(b)}(\theta_l, \theta_l) = m^{(b)}(\theta_l, \theta_l)$. $0 \leq \alpha(\theta_h, \theta_l) \leq 1$ and $0 \leq \phi(\theta_l|\theta_l) \leq 1$ holds if

$$\left(\frac{\beta_h}{\beta_l}\right)^{(a)} \leq \frac{\beta_h}{\beta_l} \leq \underbrace{\frac{((\theta_h^2 - s_h - \omega_h)(\theta_h \theta_l - s_l - \omega_l)\delta + \theta_h^2 s_l + \theta_l(s_h - s_l)\theta_h - s_h(s_l + \omega_l))(1 - \delta)}{((\theta_h \theta_l - s_l - \omega_l)\delta + s_l)(\theta_h(\theta_h - \theta_l) - s_h - \delta(\theta_h^2 - s_h - \omega_h))}}_{\equiv \left(\frac{\beta_h}{\beta_l}\right)^{(b)}}.$$

The profit is

$$\Pi_{APAM}^{(i)} = \frac{(1 - \delta)\nu(s_h)}{1 - \rho} f(\theta_h) + \frac{(1 - \delta)\nu(s_l)}{1 - \rho} f(\theta_l), \quad (34)$$

where

$$f(\theta_h) = \frac{\theta_h((\theta_l^2 - s_l - \omega_l)(\beta_h - \beta_l)\delta + (\beta_l - \beta_h)\theta_l^2 + \beta_h \theta_h \theta_l + \beta_h s_l - \beta_l(s_l + \omega_l))}{(\theta_l^2 \theta_h + (\theta_h^2 - s_h - \omega_h)\theta_l - \theta_h(s_l + \omega_l))\delta + \theta_h s_l + \theta_l s_h},$$

$$f(\theta_l) = \frac{(\beta_l - \beta_h)\theta_h \theta_l^2 + (-(\theta_h^2 - s_h - \omega_h)(\beta_h - \beta_l)\delta + \beta_h \theta_h^2 - s_h(\beta_h - \beta_l))\theta_l - \beta_l \theta_h \omega_l}{\delta \theta_h \theta_l^2 + ((\theta_h^2 - s_h - \omega_h)\delta + s_h)\theta_l - \theta_h((s_l + \omega_l)\delta - s_l)}.$$

(b.2) A_{WPAM} can be implemented by participation constraints

$$\beta_h \omega_h \leq f(\theta_h)(\delta \omega_h - (1 - \delta)s_h) + (1 - \delta)(\Phi(\theta_h, \theta_h)\theta_h^2 + \Phi(\theta_h, \theta_l)\theta_h \theta_l), \quad (35)$$

$$\beta_l \omega_l \leq f(\theta_l)(\delta \omega_l - (1 - \delta)s_l) + (1 - \delta)(\Phi(\theta_l, \theta_l)\theta_l^2 + \Phi(\theta_h, \theta_l)\theta_h \theta_l). \quad (36)$$

Feasibility and steady state constraints imply that at most one participation constraint binds if $\beta_h \omega_h \neq \beta_l \omega_l$. The steady-state masses equal inflows, $f(\theta_i) = \beta_i$, and profit is

$$\Pi_{A_{WPAM}}^{(ii)} = \frac{(1 - \delta)\nu(s_h)}{1 - \rho} \beta_h + \frac{(1 - \delta)\nu(s_l)}{1 - \rho} \beta_l. \quad (37)$$

(c) A_{NAM} : Relative to A_{WPAM} , A_{NAM} can be implemented by replacing $\beta_l \omega_l$ for $\beta_l \theta_l^2$ on the left-hand side of the constraint for the low type. This implies that more high types must be matched to low types, lowering the expected surplus for high types, and increasing rent left to low types.

(d) **Comparison:** Compare profits across the four cases: Equation 30 to 37. Set $s_h = s_l = s$. If $0 \leq s \leq \theta_l^2 - \omega_l$, the platform implements A_{WPAM} (i) if $\left(\frac{\beta_h}{\beta_l}\right)^{(a)} \leq \frac{\beta_h}{\beta_l} \leq \left(\frac{\beta_h}{\beta_l}\right)^{(b)}$. Otherwise, it implements A_{PAM} . If $\frac{\beta_h}{\beta_l} \leq \left(\frac{\beta_h}{\beta_l}\right)^{(a)}$, the platform can only implement A_{PAM} (ii) or A_{WPAM} (ii), and $\Pi_{A_{PAM}}^{(ii)} > \Pi_{A_{WPAM}}^{(ii)}$. If $\frac{\beta_h}{\beta_l} \geq \left(\frac{\beta_h}{\beta_l}\right)^{(a)}$, both A_{PAM} (i) or A_{WPAM} (i) are feasible and dominate A_{WPAM} (ii). Moreover, $\Pi_{A_{WPAM}}^{(i)} \geq \Pi_{A_{PAM}}^{(i)}$ if $\frac{\beta_h}{\beta_l} \geq \left(\frac{\beta_h}{\beta_l}\right)^{(a)}$. If $\frac{\beta_h}{\beta_l} \geq \left(\frac{\beta_h}{\beta_l}\right)^{(b)}$, the platform implements A_{PAM} (i) as A_{WPAM} (i) is not feasible.

If $\theta_l^2 - \omega_l \leq s \leq \theta_h \theta_l - \omega_l$, only A_{WPAM} (i) or (ii), or exclusion of low types is feasible. If $\beta_h \geq \beta_l$, the platform implements A_{WPAM} and the matching outcome is either weakly assortative or non-assortative. If $\beta_h < \beta_l$, the platform implements A_{WPAM} and the matching outcome is weakly assortative or only high types participate if s is large enough. Note that at $s = \theta_h \theta_l - \omega_l$, the matching outcome is non-assortative if $\beta_h \geq \beta_l$, whereas only high types participate if $\beta_h < \beta_l$.

If $\theta_h \theta_l \leq s \leq \theta_h^2$, low types do not participate and $\phi(\theta_h|\theta_h) = 1$. To conclude, there exists no $s \in [0, \theta_h \theta_l]$, where the platform implements PAM when both types participate.

□

C. SUPPLEMENTAL APPENDIX

C.1 BENCHMARKS

This section analyzes two polar cases, in which the intermediary has full information about agent's types and is able to extract the full rent from the matching output or the intermediary has no information about agent's types and must match agents at random.

Socially-Optimal Matching The first benchmark constitutes the case in which the intermediary (or a social planner) provides the socially-optimal matching under the premise that agent's types can be identified perfectly. The intermediary or social planner maximizes the sum of total matching outputs given that agents only search for one period. The matching output function is supermodular, i.e. types of both sides are complements. The socially-optimal matching is the solution to the linear program

$$\max_M \sum_{k=A,B} \sum_{\theta_j^{-k} \in \Theta^{-k}} \sum_{\theta_i^k \in \Theta^k} (\theta_i^k \theta_j^{-k} - \omega_i^k) m(\theta_i^k, \theta_j^{-k}) \quad (38)$$

subject to feasibility

$$\sum_{\theta_j^{-k} \in \Theta^{-k}} m(\theta_i^k, \theta_j^{-k}) \leq \beta_i^k, \forall \theta_i^k \in \Theta^k, \quad (39)$$

$$\sum_{\theta_i^k \in \Theta^k} m(\theta_i^k, \theta_j^{-k}) \leq \beta_j^{-k}, \forall \theta_j^{-k} \in \Theta^{-k}, \quad (40)$$

$$m(\theta_i^k, \theta_j^{-k}) \geq 0, \forall (\theta_i^k, \theta_j^{-k}) \in \Theta^k \times \Theta^{-k}. \quad (41)$$

The linear program follows the optimal assignment problem by Koopmans and Beckmann (1957) and Shapley and Shubik (1971). Both agents that form the match $(\theta_i^k, \theta_j^{-k})$ receive the output $\theta_i^k \cdot \theta_j^{-k}$.

The optimal matching rule that maximizes total match surplus follows the procedure: Starting with the highest possible type on side k , each agent is matched to the highest possible type on side $-k$. If there are not enough high types remaining on side $-k$, the algorithm proceeds in descending order of type on side $-k$ until all agents of the highest possible type on side k are matched. The process continues in descending order with the next highest type on side k , each time matching to the next available remaining types on side $-k$. Once all agents on $-k$ have been matched, any remaining agents on side k are assigned to their outside option.

Remark. If markets are fully symmetric, the socially optimal matching is $m(\theta_i^k, \theta_j^{-k}) = \beta_i^k$ if $\theta_i^k = \theta_j^{-k}$. The outcome is said to exhibit positive assortative matching.

If market sides are fully symmetric, $\beta_i^A = \beta_i^B$, the solution to the linear program is attained with $m(\theta_i^k, \theta_j^{-k}) \in \{0, \beta_i^k\}$, that is a pair is either matched with probability one or not matched. Although the linear program permits partial or fractional matching of agents, Dantzig (1963) showed that the maximum value of the objective is attained with probabilities in $\{0, 1\}$.

For symmetric populations of agents, optimality requires that no individual remains unmatched, such that the feasibility constraints must hold with equality. Otherwise, the social planner can increase welfare by assigning an unmatched agent to another unmatched agent as the value of their match is greater than zero. The objective is maximized if $m(\theta_i^k, \theta_j^{-k}) = \beta_i^k$ when $\theta_i^k = \theta_j^{-k}$ by applying the rearrangement inequality.

Random Matching The second benchmark is a random matching market. For example, if an intermediary has no information (data) about agents' types, and thus cannot condition on any observables, the intermediary's matching rule incorporates random meetings between agents. A random matching market may also reflect offline meetings between agents that are not intermediated by any platform.

A random matching market is a tuple $(\hat{\Theta}^k, f(\theta_i^k))_{k=A,B}$ with parameters (s_i^k, δ) . The analysis builds on the model of Lauermaun and Nöldeke (2014).²⁵

The total mass of agents on side k is $\bar{f}^k = \sum_{\theta_i^k \in \Theta^k} f(\theta_i^k)$. Since each agent can meet at most one agent per unit of time, the total mass of meetings is given by $\min\{\bar{f}^A, \bar{f}^B\}$. Given that meetings are random, the fraction of meetings that involve type θ_i^k on side k and type θ_j^{-k} on side $-k$ is then

$$\frac{f(\theta_i^k) f(\theta_j^{-k}) \min\{\bar{f}^k, \bar{f}^{-k}\}}{\bar{f}^k \cdot \bar{f}^{-k}}.$$

If $\bar{f}^k > \bar{f}^{-k}$, then the mass of agents on side k that meet their outside option is $\Phi(\theta_i^k, \omega_i^k) = \frac{\bar{f}^k - \bar{f}^{-k}}{\bar{f}^k}$. The probability to meet type θ_j^{-k} on side $-k$ conditional on being an agent of any type on side k is

$$\phi(\theta_j^{-k}) = \frac{f(\theta_j^{-k})}{\bar{f}^{-k}} \frac{\min\{\bar{f}^k, \bar{f}^{-k}\}}{\bar{f}^k},$$

where the probability that type θ_i^k on side k exits the search process in a match with

²⁵In contrast to Lauermaun and Nöldeke (2014), agents may face explicit search cost s_i^k in addition to δ .

type θ_j^{-k} is

$$\mu(\theta_i^k, \theta_j^{-k}) = \frac{(1 - \delta)\alpha(\theta_i^k, \theta_j^{-k})\phi(\theta_j^{-k})}{\delta + (1 - \delta)\sum_{\theta_j^{-k}} \alpha(\theta_i^k, \theta_j^{-k})\phi(\theta_j^{-k})},$$

where $\mu(\theta_i^k, \omega_i^k) = 1 - \sum_{\theta_j^{-k}} \mu(\theta_i^k, \theta_j^{-k})$ is the probability that type θ_i^k remains unmatched.

Let $(f(\theta_i^k), \alpha(\theta_i^k, \theta_j^{-k}))_{k=A,B}$ be a steady state. Then M with entries given by

$$m(\theta_i^k, \theta_j^{-k}) = \frac{\alpha(\theta_i^k, \theta_j^{-k})f(\theta_i^k)f(\theta_j^{-k})\min\{\bar{f}^k, \bar{f}^{-k}\}}{\bar{f}^k \cdot \bar{f}^{-k}}. \quad (42)$$

is the unique matching outcome induced by the steady state under random matching. Vice versa, if M is a steady state matching outcome then $f(\theta_i^k), \alpha(\theta_i^k, \theta_j^{-k})$ is given by

$$f(\theta_i^k) = \frac{\beta_i^k}{\delta} \mu(\theta_i^k, \omega_i^k), \quad (43)$$

$$\alpha(\theta_i^k, \theta_j^{-k}) = m(\theta_i^k, \theta_j^{-k}) \frac{\bar{f}^k \cdot \bar{f}^{-k}}{f(\theta_i^k)f(\theta_j^{-k})\min\{\bar{f}^k, \bar{f}^{-k}\}}, \quad (44)$$

where $\alpha(\theta_i^k, \theta_j^{-k}) \leq 1$ for all $(\theta_i^k, \theta_j^{-k}) \in \hat{\Theta}^k \times \hat{\Theta}^{-k}$ and $m(\theta_i^k, \omega_i^k)$ is the probability of ending up with one's outside option. Matching M is an **equilibrium** matching if

$$m(\theta_i^k, \theta_j^{-k}) = \begin{cases} 0 & \text{if } \theta_i^k \theta_j^{-k} < V^C(\theta_i^k) \text{ or } \theta_i^k, \theta_j^{-k} < V^C(\theta_j^{-k}) \\ \frac{f(\theta_i^k)f(\theta_j^{-k})\min\{\bar{f}^k, \bar{f}^{-k}\}}{\bar{f}^k \cdot \bar{f}^{-k}} & \text{if } \theta_i^k \theta_j^{-k} > V^C(\theta_i^k) \text{ and } \theta_i^k \theta_j^{-k} > V^C(\theta_j^{-k}) \end{cases}$$

holds for all $(\theta_i^k, \theta_j^{-k}) \in \hat{\Theta}^k \times \hat{\Theta}^{-k}$.

C.2 PROOF OF PROPOSITION 1: ADVERTISING

This addition to the proof of Proposition 1 provides the detailed mathematical steps required for completing the proof of the third case: advertisement.

Consider pairs $(\theta_1^k, \theta_A^{-k})$ and $(\theta_A^k, \theta_R^{-k})$ for which $\alpha(\theta_1^k, \theta_A^{-k}) = 1$ and $\alpha(\theta_A^k, \theta_R^{-k}) = 1$. I show that the platform's profit can be improved by changing the matching rules for types $\theta_1^k, \theta_A^k, \theta_A^{-k}$ and θ_R^{-k} as well as adjusting their search costs. The platform will choose the mass of recommended pairs $\Phi'(\theta_1^k, \theta_R^{-k}), \Phi'(\theta_1^k, \theta_A^{-k}), \Phi'(\theta_A^k, \theta_R^{-k}), \Phi'(\theta_A^k, \theta_A^{-k})$, and the mass of matched pairs $m'(\cdot, \cdot) = \alpha(\cdot, \cdot)\Phi'(\cdot, \cdot)$ as detailed below. For all other types, the platform chooses the mass of recommended pairs such that they equal the mass of recommended pairs under random matching: $\Phi'(\theta_i^k, \theta_j^{-k}) = \Phi^{RM}(\theta_i^k, \theta_j^{-k})$.

For $\varepsilon \in [-\min\{\beta_i - m^{RM}(\cdot, \cdot)\}, \min\{m^{RM}(\cdot, \cdot)\}]$, the platform chooses $m^{RM}(\theta_1^k, \theta_A^{-k}) - m'(\theta_1^k, \theta_A^{-k}) = \varepsilon$ and $m^{RM}(\theta_A^k, \theta_R^{-k}) - m'(\theta_A^k, \theta_R^{-k}) = \varepsilon$, i.e. the platform changes the mass of the two matched pairs by ε . Substituting the change into the steady state condition (Equation 10) for type θ_1^k and type θ_R^{-k} yields

$$\begin{aligned} f^{RM}(\theta_1^k) + \frac{1-\delta}{\delta}\varepsilon &= \frac{1}{\delta} \left(\beta_1^k - (1-\delta) \left(-\varepsilon + \sum_{\Theta^{-k}} m^{RM}(\theta_1^k, \theta_j^{-k}) \right) \right), \\ f^{RM}(\theta_R^{-k}) + \frac{1-\delta}{\delta}\varepsilon &= \frac{1}{\delta} \left(\beta_R^{-k} - (1-\delta) \left(-\varepsilon + \sum_{\Theta^k} m^{RM}(\theta_i^k, \theta_R^{-k}) \right) \right). \end{aligned}$$

Therefore, by decreasing (increasing) the mass of the two matched pairs, increases (decreases) the steady state mass by $\frac{1-\delta}{\delta}\varepsilon$ compared to the steady state mass under random matching. Substituting $\Phi^{RM}(\theta_1^k, \theta_A^{-k}) - \Phi'(\theta_1^k, \theta_A^{-k}) = \varepsilon$ and $\Phi^{RM}(\theta_A^k, \theta_R^{-k}) - \Phi'(\theta_A^k, \theta_R^{-k}) = \varepsilon$ into the feasibility constraints of type θ_1^k and type θ_R^{-k} yields

$$f^{RM}(\theta_1^k) + \frac{1-\delta}{\delta}\varepsilon = \Phi'(\theta_1^k, \theta_R^{-k}) - \varepsilon + \mathbf{1}_{k=A} \Phi^{RM}(\theta_1^k, \omega_1^k) + \sum_{\Theta^{-k} \setminus \{\theta_R^{-k}\}} \Phi^{RM}(\theta_1^k, \theta_j^{-k}), \quad (45)$$

$$f^{RM}(\theta_R^{-k}) + \frac{1-\delta}{\delta}\varepsilon = \Phi'(\theta_A^k, \theta_R^{-k}) - \varepsilon + \mathbf{1}_{k=A} \Phi^{RM}(\theta_R^{-k}, \omega_R^{-k}) + \sum_{\Theta^k \setminus \{\theta_1^k\}} \Phi^{RM}(\theta_i^k, \theta_R^{-k}), \quad (46)$$

which implies that $\Phi'(\theta_1^k, \theta_R^{-k}) - \Phi^{RM}(\theta_1^k, \theta_R^{-k}) = \frac{1-\delta}{\delta}\varepsilon + \varepsilon = \varepsilon/\delta$. It remains to determine $\Phi'(\theta_A^k, \theta_A^{-k}) - \Phi^{RM}(\theta_A^k, \theta_A^{-k})$ and $m'(\theta_A^k, \theta_A^{-k}) - m^{RM}(\theta_A^k, \theta_A^{-k})$. To do so, consider two cases: either $\alpha(\theta_A^k, \theta_A^{-k}) = 0$ or $\alpha(\theta_A^k, \theta_A^{-k}) = 1$.

In the first case, $\alpha(\theta_A^k, \theta_A^{-k}) = 0$, I can exchange θ_A^k for θ_1^k and θ_A^{-k} for θ_R^{-k} in Equation 45 and 46 above. Then, it follows that $\Phi'(\theta_A^k, \theta_A^{-k}) - \Phi^{RM}(\theta_A^k, \theta_A^{-k}) = \varepsilon/\delta$ and $m'(\theta_A^k, \theta_A^{-k}) = 0$.

In the second case, $\alpha(\theta_A^k, \theta_A^{-k}) = 1$, the platform can set $\Phi'(\theta_A^k, \theta_A^{-k}) - \Phi^{RM}(\theta_A^k, \theta_A^{-k}) = m'(\theta_A^k, \theta_A^{-k}) - m^{RM}(\theta_A^k, \theta_A^{-k}) = \varepsilon$. Since the platform decreases (increases) the mass of the matched pair $(\theta_A^k, \theta_R^{-k})$ but increases (decreases) the mass of the matched pair $(\theta_A^k, \theta_A^{-k})$ by the same amount, this implies that the steady state mass of type θ_A^k is unchanged compared to the steady state masses under random matching. Additionally, feasibility continues to be satisfied as the platform shift mass ε from one recommended pair to the other. Similarly, the steady state mass of type θ_A^{-k} is the same as under random matching and the steady state constraint as well as feasibility constraint remain satisfied.

Next determine the change in search costs for types $\theta_1^k, \theta_A^k, \theta_A^{-k}$ and θ_R^{-k} . Note that for the newly chosen mass of recommended and matched pairs $(\Phi'(\cdot, \cdot), m'(\cdot, \cdot))$, the originally binding participation or incentive constraint is no longer binding. Since, however, the right-hand side of the participation or incentive constraints (see Equation 22 and 23) are linearly increasing in $m(\cdot, \cdot)$ and the left-hand side of the constraints are ordered due to the supermodularity of the match utility, the platform can choose a new search cost $\tilde{s}_i^k$ such that the constraint becomes binding again. Let the platform choose $\tilde{s}_1^k, \tilde{s}_A^k, \tilde{s}_R^{-k}, \tilde{s}_A^{-k}$ such that originally binding participation or incentive constraint of each type is binding again. Using Equations 22 and 23 and $m^{RM}(\theta_1^k, \theta_A^{-k}) - m'(\theta_1^k, \theta_A^{-k}) = \varepsilon$ and $m^{RM}(\theta_A^k, \theta_R^{-k}) - m'(\theta_A^k, \theta_R^{-k}) = \varepsilon$, the difference between $s_1^k - \tilde{s}_1^k$ and $s_R^{-k} - \tilde{s}_R^{-k}$ is given by

$$(1 - \delta) \frac{\beta_1^k - (1 - \delta) \sum_j m^{RM}(\theta_1^k, \theta_j^{-k})}{\delta} (s_1^k - \tilde{s}_1^k) = \varepsilon (\theta_1^k \theta_A^{-k} - \omega_1^k + \frac{1 - \delta}{\delta} \tilde{s}_1^k). \quad (47)$$

Observe that the right-hand side is positive for $\varepsilon > 0$ as $\theta_1^k \theta_A^{-k} - \omega_1^k > 0$ due to the fact that both types mutually accept each other. Then, it follows that $\tilde{s}_1^k$ must be smaller than s_1^k for $\varepsilon > 0$. Additionally, by the steady state constraint, the factor on the left-hand side is equal to $(1 - \delta) f^{RM}(\theta_1^k)$. Similarly, using $m^{RM}(\theta_A^k, \theta_R^{-k}) - m'(\theta_A^k, \theta_R^{-k}) = \varepsilon$ and taking the difference, $s_R^{-k} - \tilde{s}_R^{-k}$ is given by

$$(1 - \delta) f^{RM}(\theta_R^{-k}) (s_R^{-k} - \tilde{s}_R^{-k}) = \varepsilon (\theta_A^k \theta_R^{-k} - \omega_R^{-k} + \frac{1 - \delta}{\delta} \tilde{s}_R^{-k}). \quad (48)$$

Observe that the right-hand side is again positive, so that $s_R^{-k} > \tilde{s}_R^{-k}$ for $\varepsilon > 0$. Next, if $\alpha(\theta_A^k, \theta_A^{-k}) = 0$, the difference between the search costs for types θ_A^k and θ_A^{-k} can be derived as above

$$(1 - \delta) f^{RM}(\theta_A^k) (s_A^k - \tilde{s}_A^k) = \varepsilon (\theta_A^k \theta_R^{-k} - \omega_A^k + \frac{1 - \delta}{\delta} \tilde{s}_A^k), \quad (49)$$

$$(1 - \delta) f^{RM}(\theta_A^{-k}) (s_A^{-k} - \tilde{s}_A^{-k}) = \varepsilon (\theta_1^k \theta_A^{-k} - \omega_A^{-k} + \frac{1 - \delta}{\delta} \tilde{s}_A^{-k}). \quad (50)$$

Again, it holds that $s_A^k > \tilde{s}_A^k$ and $s_A^{-k} > \tilde{s}_A^{-k}$ for $\varepsilon > 0$. If $\alpha(\theta_A^k, \theta_A^{-k}) = 1$, recall that the platform sets: $m^{RM}(\theta_1^k, \theta_A^{-k}) - m'(\theta_1^k, \theta_i^{-k}) = \varepsilon$ $m^{RM}(\theta_A^k, \theta_R^{-k}) - m'(\theta_A^k, \theta_R^{-k}) = \varepsilon$ and

$m^{RM}(\theta_A^k, \theta_A^{-k}) - m'(\theta_A^k, \theta_A^{-k}) = -\varepsilon$. Using Equations 22 and 23, the difference of $s_A^{-k} - \tilde{s}_A^{-k}$ is given by

$$(1 - \delta)f^{RM}(\theta_A^{-k})(s_A^{-k} - \tilde{s}_A^{-k}) = \varepsilon(\theta_A^k \theta_A^{-k} - \omega_A^{-k} + \frac{1 - \delta}{\delta} \tilde{s}_A^{-k}) - \varepsilon(\theta_A^k \theta_A^{-k} - \omega_A^{-k} + \frac{1 - \delta}{\delta} \tilde{s}_A^{-k}). \quad (51)$$

Since $\theta_A^k > \theta_1^k$, the right-hand side is negative, so that $s_A^{-k} < \tilde{s}_A^{-k}$ for $\varepsilon > 0$. Similarly, the difference of $s_A^k - \tilde{s}_A^k$ is given by

$$(1 - \delta)f^{RM}(\theta_A^k)(s_A^k - \tilde{s}_A^k) = \varepsilon(\theta_A^k \theta_R^{-k} - \theta_A^k \theta_A^{-k}), \quad (52)$$

where the right-hand side is non-negative if $\theta_R^{-k} \geq \theta_A^{-k}$.

To determine whether the deviation is profitable, consider the difference in profits between the deviation profits and random matching profits (Equation 24). In the first case, when $\alpha(\theta_A^k, \theta_A^{-k}) = 0$, the steady state mass of all four types $(\theta_1^k, \theta_A^k, \theta_R^{-k}, \theta_A^{-k})$ increases (decreases) by $(1 - \delta)\varepsilon/\delta$ while their search costs decrease (increase). The difference in profits is therefore

$$\sum_{\theta_i^k \in \{\theta_1^k, \theta_A^k, \theta_R^{-k}, \theta_A^{-k}\}} \left[(\nu(\tilde{s}_i^k) - \nu(s_i^k))f^{RM}(\theta_i^k) + \nu(\tilde{s}_i^k) \frac{(1 - \delta)\varepsilon}{\delta} \right].$$

Differentiating with respect to ε_i^k , and evaluating the condition at $\varepsilon = 0$, yields:

$$\sum_{\theta_i^k \in \{\theta_1^k, \theta_A^k, \theta_R^{-k}, \theta_A^{-k}\}} \nu'(\tilde{s}_i^k) \Big|_{\tilde{s}_i^k = s_i^k} \frac{\partial \tilde{s}_i^k}{\partial \varepsilon} \Big|_{\varepsilon=0} f^{RM}(\theta_i^k) + \frac{1 - \delta}{\delta} \nu(\tilde{s}_i^k) \Big|_{\tilde{s}_i^k = s_i^k}. \quad (53)$$

Totally differentiating Equations 47, 48, 49, and 50 and evaluating the derivative at $\varepsilon = 0$ yields:

$$\begin{aligned} \frac{\partial \tilde{s}_1^k}{\partial \varepsilon} \Big|_{\varepsilon=0} &= -\frac{\theta_1^k \theta_A^{-k} - \omega_1^k + \frac{1 - \delta}{\delta} s_1^k}{(1 - \delta)f^{RM}(\theta_1^k)}, & \frac{\partial \tilde{s}_A^k}{\partial \varepsilon} \Big|_{\varepsilon=0} &= -\frac{\theta_A^k \theta_R^{-k} - \omega_A^k + \frac{1 - \delta}{\delta} s_A^k}{(1 - \delta)f^{RM}(\theta_A^k)}, \\ \frac{\partial \tilde{s}_R^{-k}}{\partial \varepsilon} \Big|_{\varepsilon=0} &= -\frac{\theta_A^k \theta_R^{-k} - \omega_R^{-k} + \frac{1 - \delta}{\delta} s_R^{-k}}{(1 - \delta)f^{RM}(\theta_R^{-k})}, & \frac{\partial \tilde{s}_A^{-k}}{\partial \varepsilon} \Big|_{\varepsilon=0} &= -\frac{\theta_1^k \theta_A^{-k} - \omega_A^{-k} + \frac{1 - \delta}{\delta} s_A^{-k}}{(1 - \delta)f^{RM}(\theta_A^{-k})}. \end{aligned}$$

For analytical convenience, consider the class of concave functions $\nu(s_i^k) = \kappa(s_i^k)^\alpha$ for $\kappa \in \mathbb{R}^+$ and $\alpha \in (0, 1)$ from now on. Substituting $\nu(s_i^k) = \kappa(s_i^k)^\alpha$, $\nu'(s_i^k) = \kappa\alpha(s_i^k)^{\alpha-1}$,

and the partial derivatives above into Equation 25 yields

$$\begin{aligned}
D \equiv & \underbrace{\alpha(s_1^k)^{\alpha-1} \left(-\frac{\theta_1^k \theta_A^{-k} - \omega_1^k + \frac{1-\delta}{\delta} s_1^k}{1-\delta} \right) + \frac{1-\delta}{\delta} (s_1^k)^\alpha}_{=d_1(\theta_1^k)} \\
& + \underbrace{\alpha(s_A^k)^{\alpha-1} \left(-\frac{\theta_A^k \theta_R^{-k} - \omega_A^k + \frac{1-\delta}{\delta} s_A^k}{1-\delta} \right) + \frac{1-\delta}{\delta} (s_A^k)^\alpha}_{=d_1(\theta_A^k)} \\
& + \underbrace{\alpha(s_A^{-k})^{\alpha-1} \left(-\frac{\theta_1^k \theta_A^{-k} - \omega_A^{-k} + \frac{1-\delta}{\delta} s_A^{-k}}{1-\delta} \right) + \frac{1-\delta}{\delta} (s_A^{-k})^\alpha}_{=d_1(\theta_A^{-k})} \\
& + \underbrace{\alpha(s_R^{-k})^{\alpha-1} \left(-\frac{\theta_A^k \theta_R^{-k} - \omega_R^{-k} + \frac{1-\delta}{\delta} s_R^{-k}}{1-\delta} \right) + \frac{1-\delta}{\delta} (s_R^{-k})^\alpha}_{=d_1(\theta_R^{-k})}.
\end{aligned}$$

For the deviation to be profitable, the expression must be non-zero when being evaluated at $\varepsilon = 0$. I will argue that D has at most one root in α . To do so, examine the term for θ_1^k . Differentiating with respect to α results in

$$\frac{\partial d_1(\theta_1^k)}{\partial \alpha} = (s_1^k)^{\alpha-1} \left(-\frac{\theta_1^k \theta_A^{-k} - \omega_1^k + \frac{1-\delta}{\delta} s_1^k}{(1-\delta)} (1 + \alpha \ln(s_1^k)) + \frac{1-\delta}{\delta} s_1^k \ln(s_1^k) \right).$$

Now, observe that $(s_1^k)^{\alpha-1}$ is strictly increasing in α , whereas the expression in brackets changes sign at most once since it is linear in α . This implies that $\frac{\partial d_1(\theta_1^k)}{\partial \alpha}$ changes sign at most once, in which case it is positive for some $\alpha < \alpha'$ and negative for $\alpha > \alpha'$. Similarly, this holds for the equivalent expressions, $d_1(\cdot)$, for each type $\theta_A^k, \theta_A^{-k}, \theta_R^{-k}$. Then, since the function D is continuous in α , $D > 0$ for $\alpha = 0$, and D is increasing in α for $\alpha < \alpha''$ and decreasing for $\alpha > \alpha''$, it follows that D has at most one root.

In the second case, when $\alpha(\theta_A^k, \theta_A^{-k}) = 1$, the steady state mass of types $(\theta_1^k, \theta_R^{-k})$ increases (decreases) by $\frac{(1-\delta)\varepsilon}{\delta}$ while the steady state mass of types θ_A^k and θ_A^{-k} remains unchanged. The difference in profits is therefore

$$\sum_{\theta_i^k \in \{\theta_1^k, \theta_R^{-k}\}} \left[(\nu(\tilde{s}_i^k) - \nu(s_i^k)) f^{RM}(\theta_i^k) + \nu(\tilde{s}_i^k) \frac{1-\delta}{\delta} \varepsilon \right] + \sum_{\theta_i^k \in \{\theta_A^k, \theta_A^{-k}\}} [(\nu(\tilde{s}_i^k) - \nu(s_i^k)) f^{RM}(\theta_i^k)].$$

Differentiating with respect to ε and evaluating the condition at $\varepsilon = 0$, yields

$$\sum_{\theta_i^k \in \{\theta_1^k, \theta_R^{-k}\}} \left[\nu'(\tilde{s}_i^k) \Big|_{\tilde{s}_i^k = s_i^k} \frac{\partial \tilde{s}_i^k}{\partial \varepsilon} \Big|_{\varepsilon=0} f^{RM}(\theta_i^k) + \frac{1-\delta}{\delta} \nu(\tilde{s}_i^k) \Big|_{\tilde{s}_i^k = s_i^k} \right] \quad (54)$$

$$+ \sum_{\theta_i^k \in \{\theta_A^k, \theta_A^{-k}\}} \left[\nu'(\tilde{s}_i^k) \Big|_{\tilde{s}_i^k = s_i^k} \frac{\partial \tilde{s}_i^k}{\partial \varepsilon} \Big|_{\varepsilon=0} f^{RM}(\theta_i^k) \right]. \quad (55)$$

Again, the expression must be non-zero for the deviation to be profitable. Again, I totally differentiate Equations 51 and 52:

$$\frac{\partial \tilde{s}_A^k}{\partial \varepsilon} \Big|_{\varepsilon=0} = \frac{\theta_A^k(\theta_A^{-k} - \theta_R^{-k})}{(1-\delta)f^{RM}(\theta_A^k)} > 0 \text{ if } \theta_A^{-k} > \theta_R^{-k}, \quad \frac{\partial \tilde{s}_A^{-k}}{\partial \varepsilon} \Big|_{\varepsilon=0} = \frac{\theta_A^{-k}(\theta_A^k - \theta_1^k)}{(1-\delta)f^{RM}(\theta_A^{-k})} > 0.$$

Substituting $\nu(s_i^k) = \kappa(s_i^k)^\alpha$, $\nu'(s_i^k) = \kappa\alpha(s_i^k)^{\alpha-1}$, and the partial derivatives above into Equation 55 yields

$$D_2 \equiv d_1(\theta_1^k) + \underbrace{\alpha(s_A^k)^{\alpha-1} \left(\frac{\theta_A^k(\theta_A^{-k} - \theta_R^{-k})}{(1-\delta)} \right)}_{d_2(\theta_A^k)} + \underbrace{\alpha(s_A^{-k})^{\alpha-1} \left(\frac{\theta_A^{-k}(\theta_A^k - \theta_1^k)}{(1-\delta)} \right)}_{=d_2(\theta_A^{-k})} + d_1(\theta_R^{-k}).$$

Examining the two new terms shows that $d_2(\theta_A^{-k})$ is strictly increasing in α , and $d_2(\theta_A^k)$ is strictly increasing in α if $\theta_A^{-k} > \theta_R^{-k}$, and decreasing otherwise. Again, this implies that D_2 has at most one root.

Now, suppose $\beta_i^k, \omega_i^k, \theta_i^k, \theta_j^{-k}$ and δ are drawn uniformly from their continuous intervals. Then, there exists at most one α for which $D = 0$ (or $D_2 = 0$). Let α be drawn uniformly from $(0, 1)$, then random matching is generically suboptimal as such α is drawn with measure zero. $\square$